

Serviceability Based Design of Shotcrete with Fiber Reinforced Concrete

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1st International Symposium on Fiber Reinforced Shotcrete for Underground Mining (InFUM)

- <https://infum.com.br/>
- October 1, 2023 - October 4, 2023
- Pontifical Catholic University of Rio de Janeiro (PUC-Rio), Brazil
- underground mining industry, research community, and all related sectors to share knowledge and practical experience.
- Keynote speakers, panel discussions, and workshops focused on the latest trends in the underground mining industry. Networking

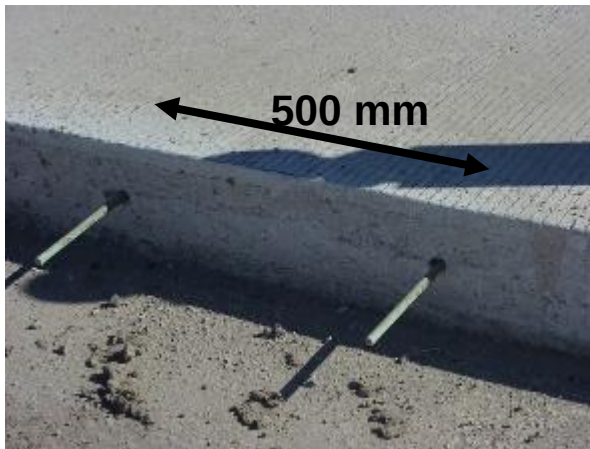


Outline

- Objectives
- Overview of FRC material responses
- Parametric Material models
- Algorithm
- Closed form solutions for moment curvature diagram
- Load deflection response
- Simulations
- Conclusions

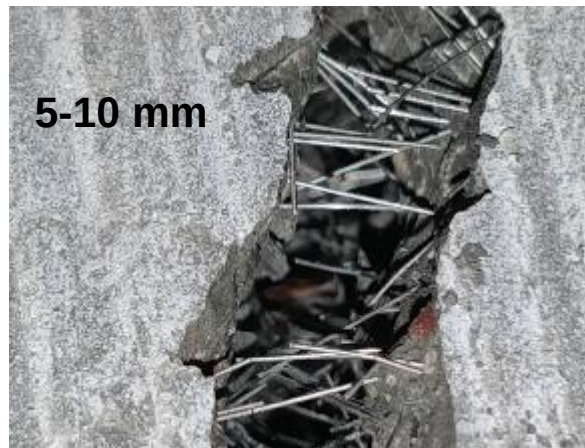
Why Should we move towards Hybrid reinforcement strategies?

Strengthening at the microstructure level will have a profound impact at the macro-structural level strength and durability



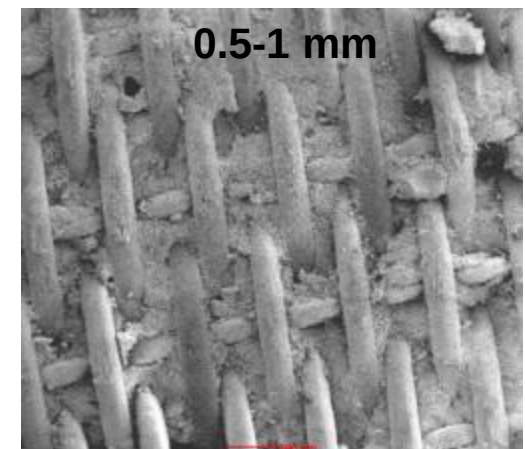
Ordinary reinforced concrete slab, 5-10" thick, standard rebar, 12"-18" apart

Reinforcing Bars:
Structural reinforcement



Fiber reinforced Concrete (FRC), reinforced at the meso-structure with $\frac{1}{2}$ " long fibers, $\frac{1}{2}$ " (10mm) millimeters apart

Fiber reinforced Concrete: Meso-structure reinforcement



Textile reinforced Concrete (TRC), reinforced at the micro-structure with long fibers, woven, and only millimeters apart

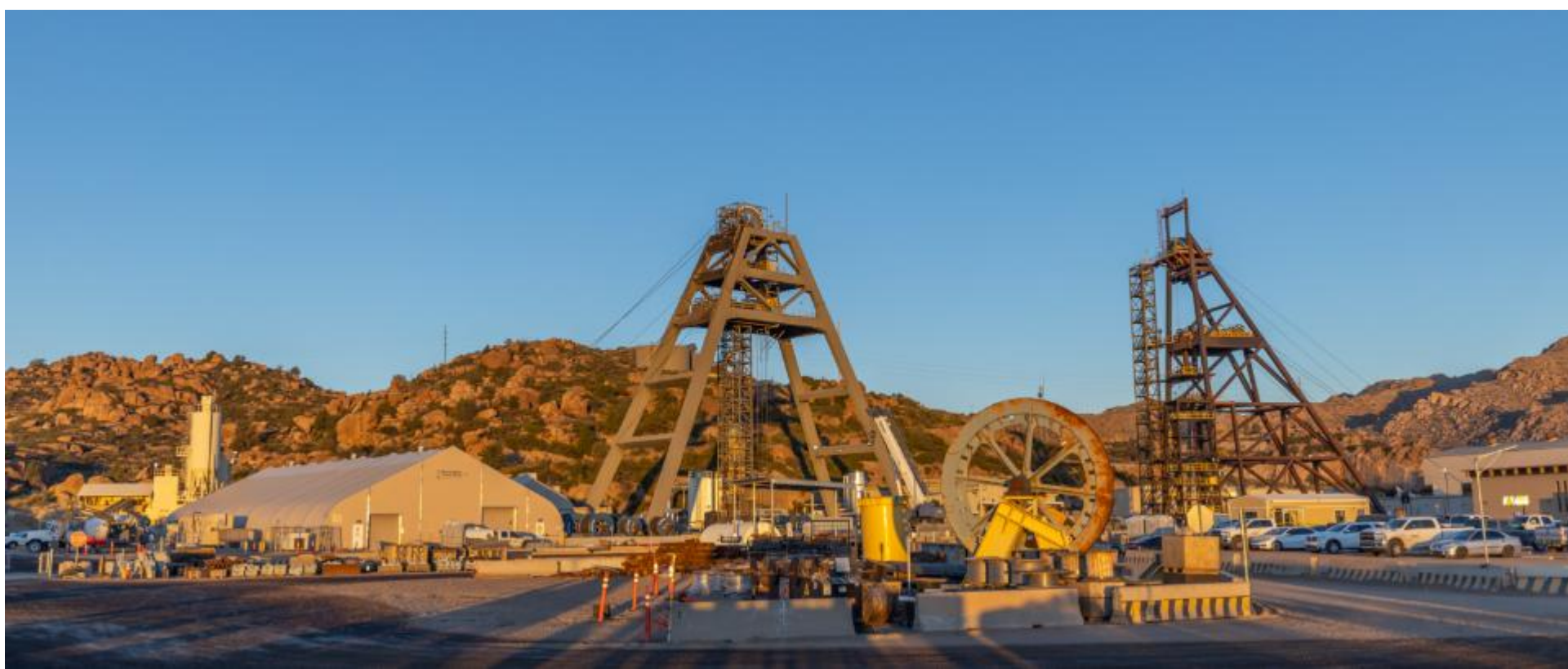
Textile Reinforced Concrete: Micro-structure reinforcement

Sustainability of Construction Materials

- At about 20,000,000,000 tons per year, Concrete is the 2nd consumed material per capita in the world. (after water)
- \$1.6 trillion is needed over a 5-year period to bring the US's infrastructure to a good condition. (ASCE, 2009)
- How do we address sustainability?
 - Efficient use of resources in concrete production such as use of by-products, less cement consumption, less CO₂ production
 - Novel construction techniques and materials, low-energy and long-lasting concrete structures.
 - Development of Design and Analysis tools to address service life.

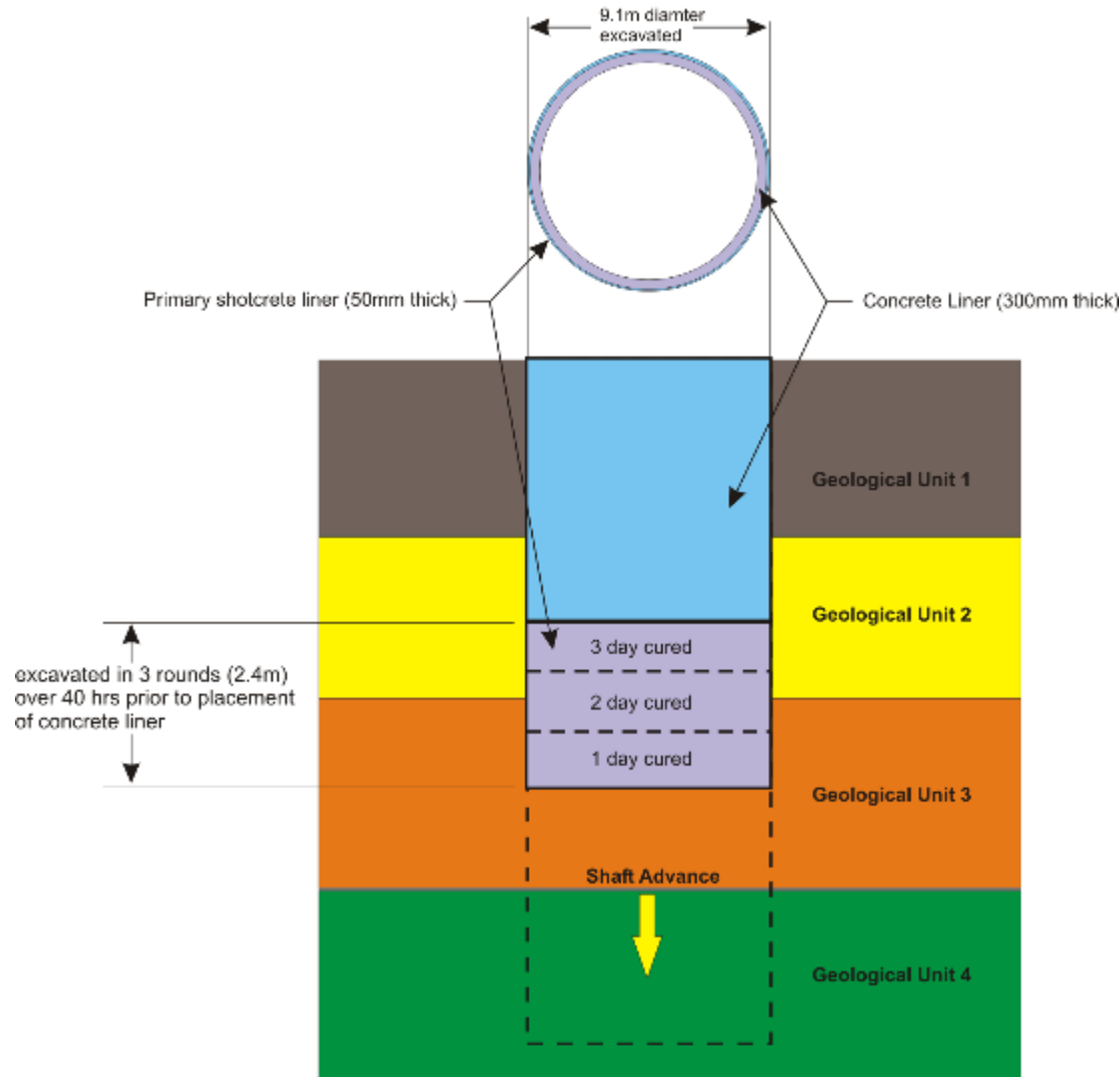
Resolution Copper – Magma Copper mine, Superior, Arizona

ASU evaluated the initial fiber-reinforced shotcrete support design for a 2000 m-deep shaft at Resolution Mine, Arizona.



Project Overview

- Evaluate fiber-reinforced shotcrete performance
- No rockbolts planned for initial support.
- Deep shaft (2189 m)
- Three geological units
- A range of stress conditions (increased with depth)
- Several modes of instability: gravity driven, rockmass shear yielding, brittle failure



Shaft used in the feasibility study at Magma Mine



Shotcrete is applied prior to form setting for permanent lining



Traditional Rock Bolt Mesh System is time consuming



Resolution Copper mine

- The Resolution Copper deposit ranges from **1,500 to 2,130 m** below the surface.
- Average grade is estimated at **1.5% copper**.
- Potentially the largest copper mine in North America, producing up to **25% percent of U.S. copper demand** per year.
- After extensive construction and ramp-up period, ore production is approximately **120,000 tons per day**
- Estimated to produce as much as **40 billion pounds of copper** over 40 years.
- **Shotcrete lining system** as initial shaft support instead of the traditional mesh/bolt increasing the development speed of the shaft.
- Considered aspects: alternatives to achieve adhesion, strength, shotcrete ductility.
- Material innovativeness, analysis, design methodology, and the constructability.
- The shotcrete system must achieve a **high early strength and ductility within a short period (less than 24 hours)**.

Geotechnical Conditions

Geological Domains along shaft

Unit	Rock Type	Depth (m)	Thickness (m)
Apache Leap (Tal)	Volcanic Tuff	0-520	520
Whitetail (TW-cgl/TW-ss)	Conglomerate/sandstone	520-920	400
KVS (KVS-phy)	Volcanic sediments	920-1835	915

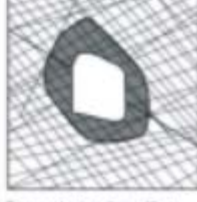
In-Situ Stress Domains at Resolution Mine

Stress Domain		Stress Component	Magnitude: Base+Gradient (MPa)	Azimuth (true north)
Geological Units	Approximate Depths (m)			
TVols: Apache Leap & Whitetail Conglomerate	0-1000	SHmax	10+0.030D	North-South
		Shmin	5+0.013D	East-Weat
		Svert	0+0.024D	vertical
IASeds: below whitetail conformity	1000-2500	SHmax	32+0.0080D	North-South
		Shmin	12.1+0.006D	East-Weat
		Svert	-3.5+0.028D	vertical

***Geotechnical Data from Hudson, 2008 and Golder, 2006**

Note: "D" is depth in meters below surface.

Analysis Methods

	Massive ($RMR > 75$)	Moderately Fractured ($50 > RMR > 75$)	Highly Fractured ($RMR < 50$)	
Low In-Situ Stress ($\sigma_1 / \sigma_c < 0.15$)	 Linear elastic response.	 Falling or sliding of blocks and wedges.	 Unraveling of blocks from the excavation surface.	Low Mining-Induced Stress $\sigma_{max}/\sigma_c < 0.4 \pm 0.1$
Intermediate In-Situ Stress ($0.15 > \sigma_1 / \sigma_c < 0.4$)	 Brittle failure adjacent to excavation boundary.	 Localized brittle failure of intact rock and movement of blocks.	 Localized brittle failure of intact rock and unraveling along discontinuities.	Intermediate Induced Stress $0.4 \pm 0.1 < \sigma_{max}/\sigma_c < 1.15 \pm 0.1$
High In-Situ Stress ($\sigma_1 / \sigma_c > 0.4$)	 Brittle failure around the excavation.	 Brittle failure of intact rock around the excavation and movement of blocks.	 Squeezing and swelling rocks. Elastic/plastic continuum.	High Mining-Induced Stress $\sigma_{max}/\sigma_c > 1.15 \pm 0.1$

KVS: $GSI=60$, $\sigma_{max}/\sigma_c=1.5$

TW: $GSI=60$, $\sigma_{max}/\sigma_c=2.4$

Selection of analysis methodology (Diederichs, 2007)

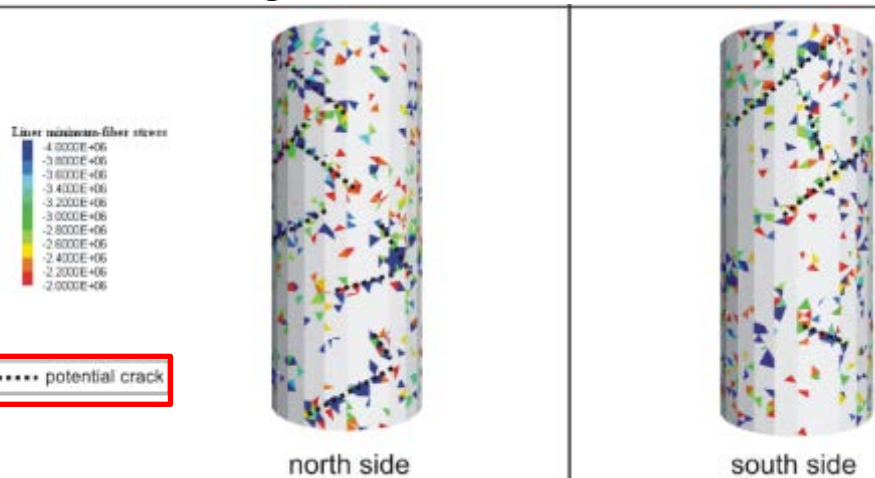
UCS/T or mi	$GSI < 55$	$GSI = 55-65$	$GSI = 65-80$	$GSI > 80$
< 9	<i>GSI</i>	<i>GSI (TW)</i>	<i>GSI</i>	<i>GSI</i>
9-15	<i>GSI</i>	<i>GSI (KVS)</i>	<i>GSI</i>	<i>GSI/DISL</i>
15-20	<i>GSI</i>	<i>GSI/DISL</i>	<i>GSI/DISL (TAL)</i>	<i>DISL</i>
> 20	<i>GSI</i>	<i>GSI/DISL</i>	<i>DISL</i>	<i>DISL</i>

Unit	Model of Instability
TAL:	Low stress – gravity driven instability Medium stress – local brittle failure
KVS:	GSI rockmass yielding and/or some local brittle failure
TW:	GSI rockmass yielding – raveling of conglomerate

Rock mass quality in geological strength index (GSI) chart

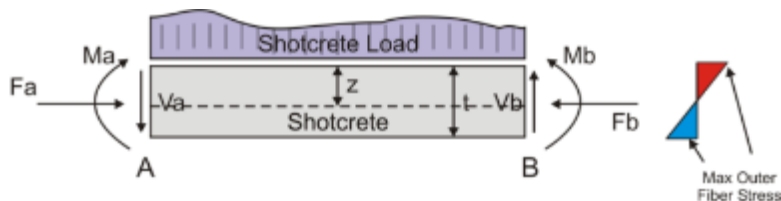
3DEC Evaluation of Shotcrete – TAL Unit

Tensile extreme fiber stress *exceeding* tensile strength of shotcrete



3DEC-predicted liner stresses and potential cracking in TAL unit

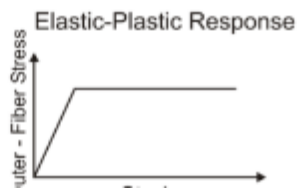
- Linear-elastic shotcrete liner installed in low stress, blocky ground.
- Early age properties use for shotcrete and rock-shotcrete interface along full length of model shaft.
- Extreme fiber stress used to evaluate potential shotcrete cracking.
- Tensile extreme fiber stress exceeding the tensile strength *combined* with tensile thrust at a given liner section indicates potential for a crack to form through the liner.
- Large regions of connected cracks would indicate potential for a block breaking through the shotcrete.



Combines Flexural Strength:

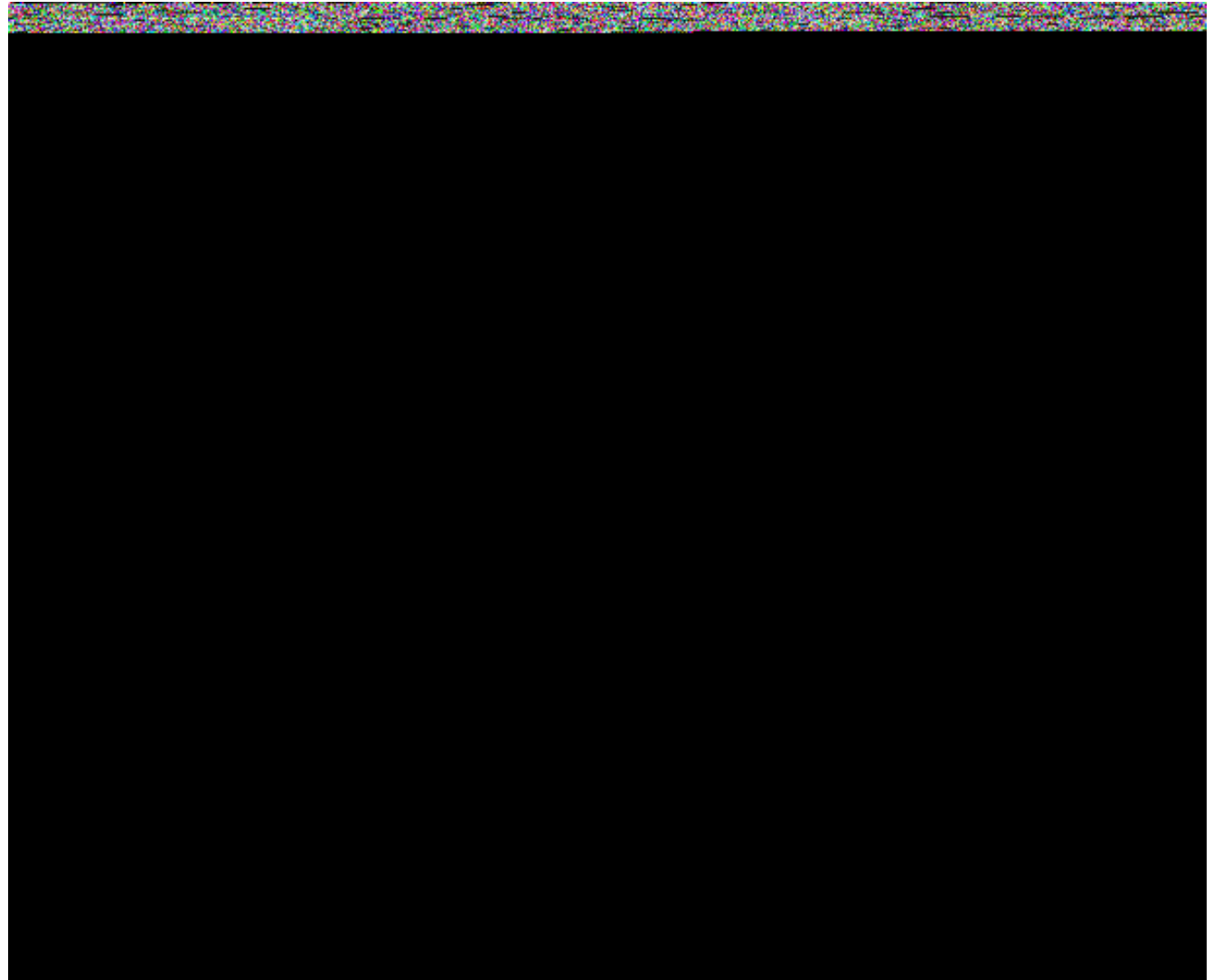
$$F/A \pm Mz/I \leq \sigma_{\text{ten, comp strength}}$$

$$A = 1 \cdot t; I = 1 \cdot t^3/12$$

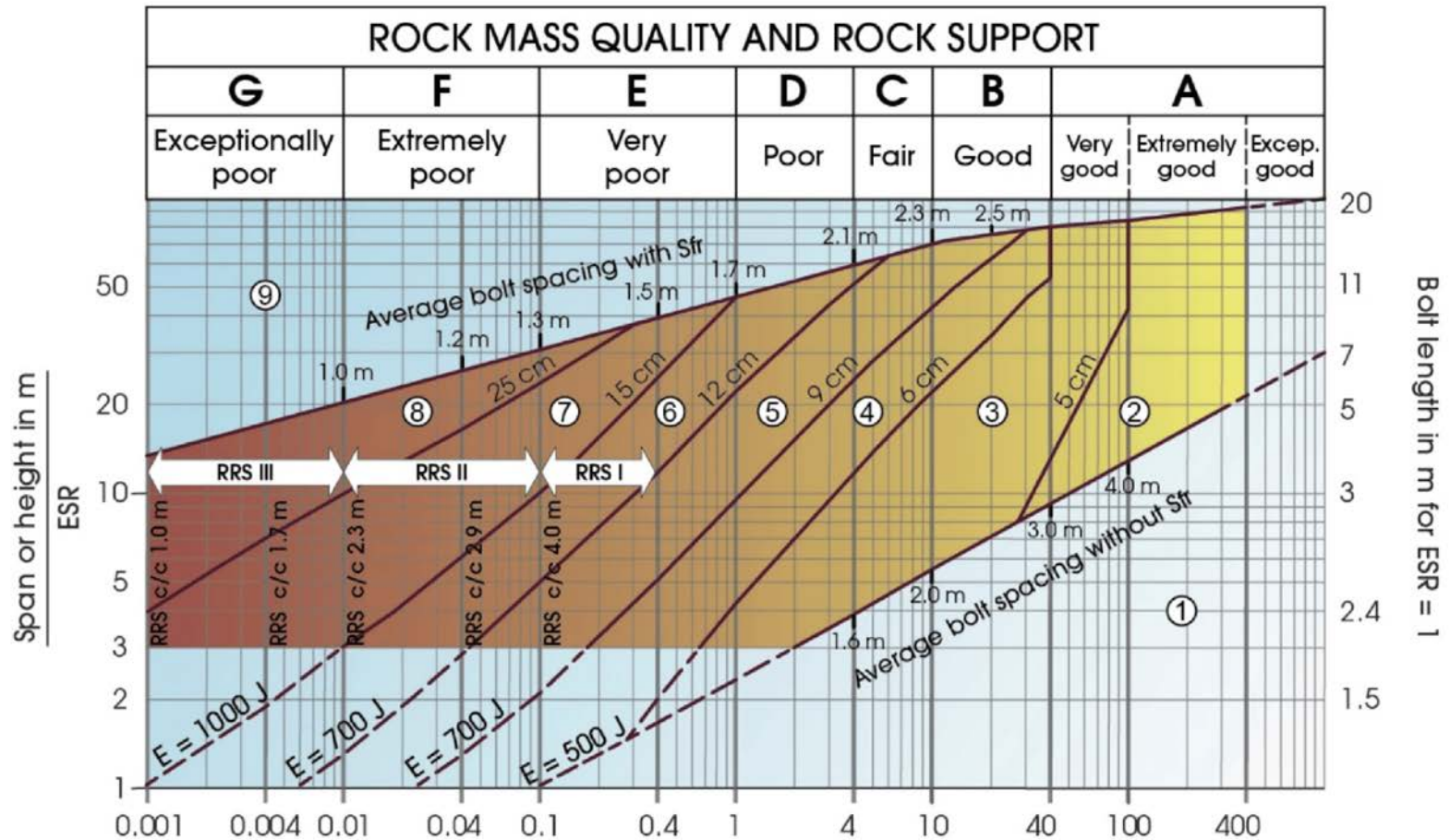


Extreme Fiber Stress

A 3DEC Numerical Analysis of the Interaction Between an Uneven Rock Surface and Shotcrete Lining

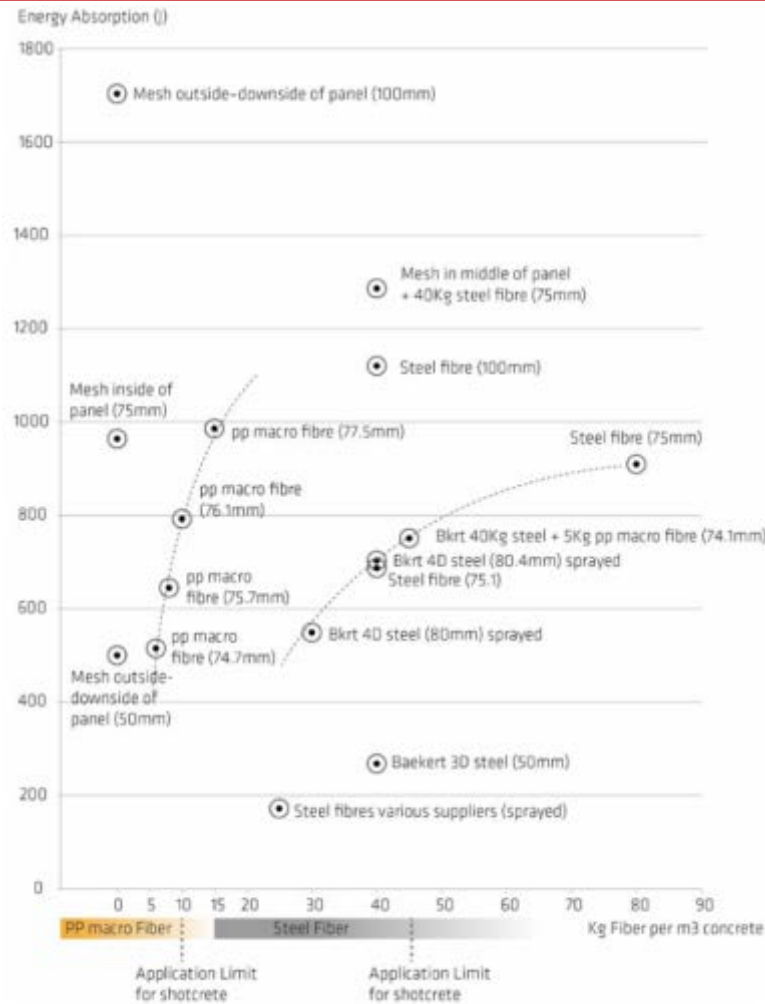


Modified Barton chart



$$\text{Rock mass quality } Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

Applications of fibers in Shotcrete is primarily quasi-scientific



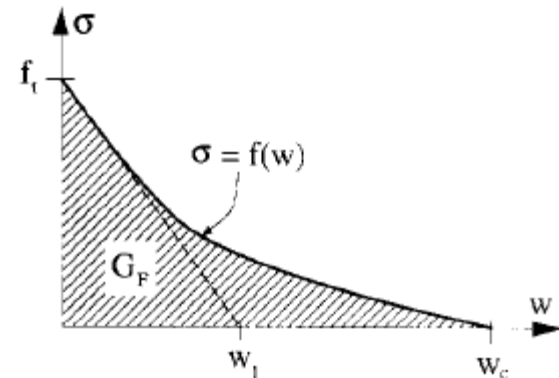
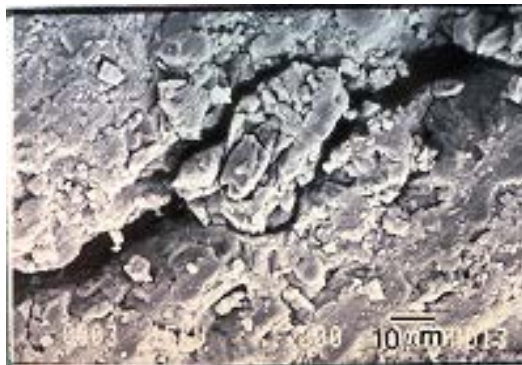
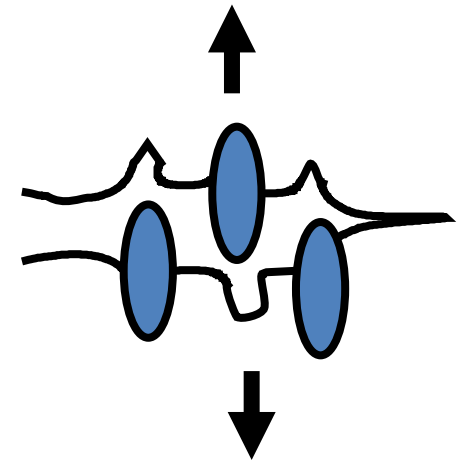
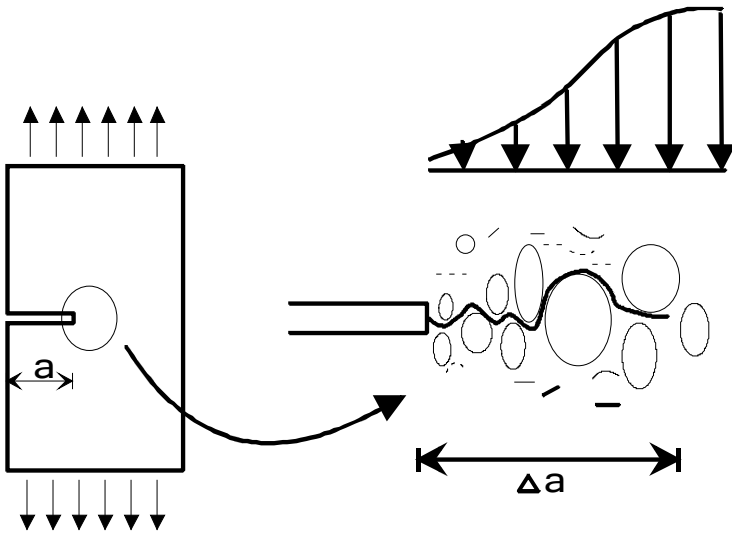
- Erismann, F & Hansson, M 2019, 'Early strength development of shotcrete for rapid mine development and behaviour under dynamic loads', in J Hadjigeorgiou & M Hudyma (eds), Ground Support 2019: Proceedings of the Ninth International Symposium on Ground Support in Mining and Underground Construction, Australian Centre for Geomechanics, Perth, pp. 559-570, https://doi.org/10.36487/ACG_rep/1925_40_Erismann

Outline of Tasks

- Three-point and Four-point bending test set up
- Back-calculation procedure
- Round panel test
- Correlation between 3PB, 4PB, and round panel test
- Elastic beam with nonlinear cracked hinged model
- Rigid crack model of round panel
- Virtual testing with finite element software ABAQUS
- Full scale testing and simulation of flat slab

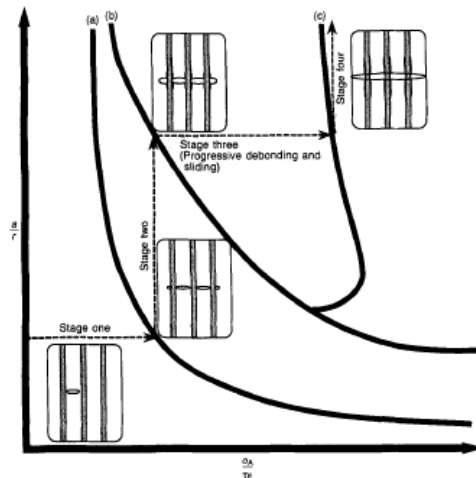
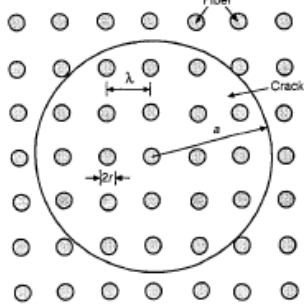
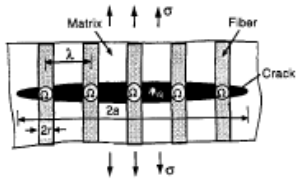
Fracture Process in Quasi-Brittle Materials

1. Stable Crack growth
2. Fracture Mechanics is applicable
3. Energy loss as the crack grows, dissipation along the crack path

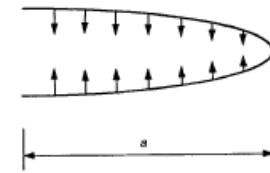
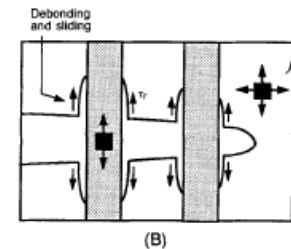
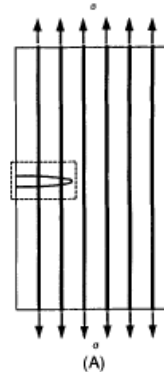
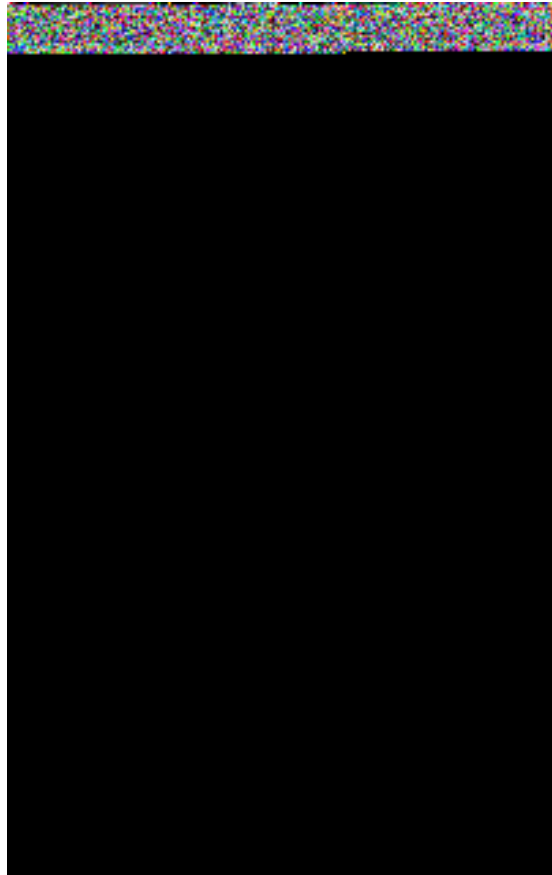


Micromechanics of toughening

why Fibers increase the tensile strength and ductility?



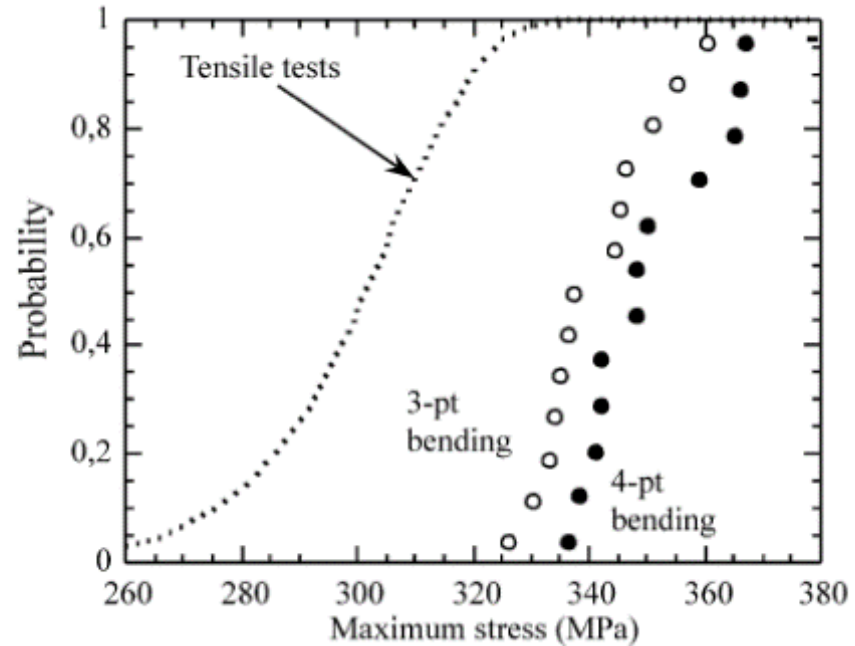
- Crack Bridging mechanisms
- Three parameters of “Stress”, “Crack opening”, “Crack length” are important tools of modeling,
- $\sigma-\omega-a$, or $\sigma(\omega)$, $\sigma(a)$, $\omega(a)$



Crack Bridging Mechanism



Role of fiber in crack bridging mechanism

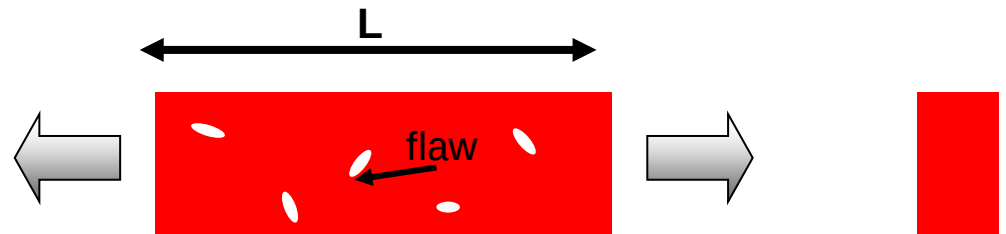


Size Effect-Fundamental differences in tension notched or unnotched, and flexural test

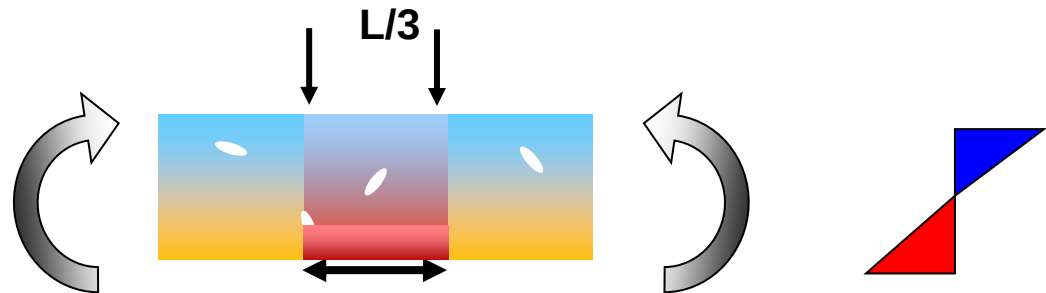
- V.Calard, J.Lamon, "A probabilistic-statistical approach to the ultimate failure of ceramic-matrix composites—part I: experimental investigation of 2D woven SiC/SiC composites," , Composites Science and Technology 62 (2002) 385–393

Uniaxial Tension vs. Flexural Bending

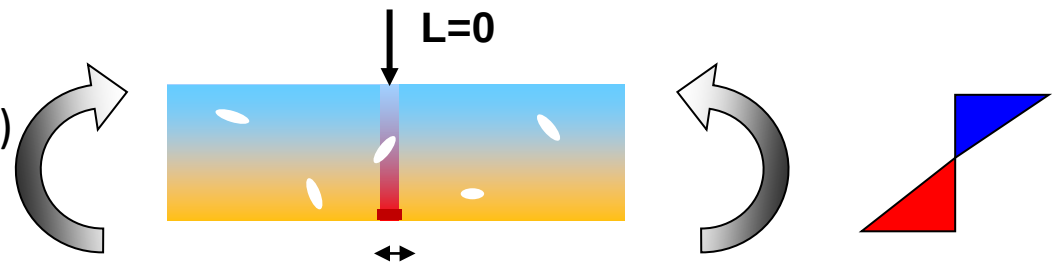
- Uniaxial Tension



- Flexural Bending (4 point bending)



- Flexural Bending (3 point bending)



Shotcrete for repair of canal lining





Current Shotcreting method is an extremely labor-intensive process

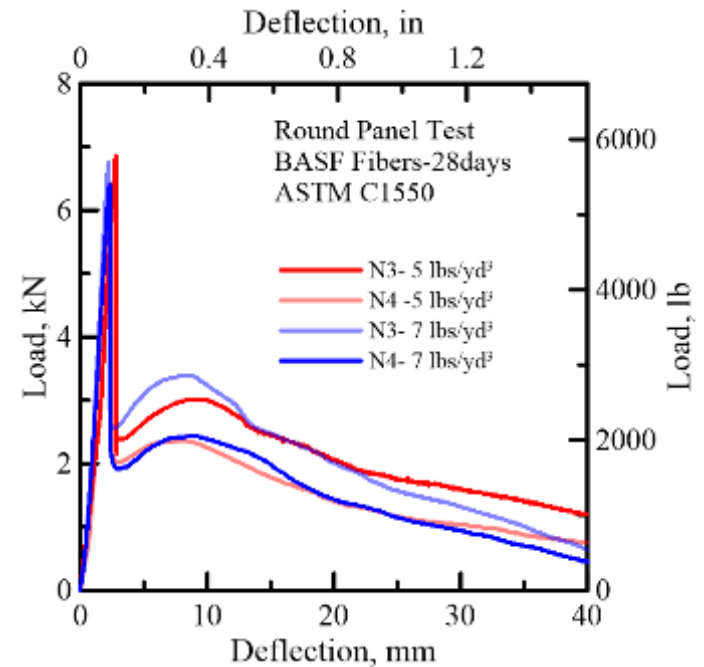


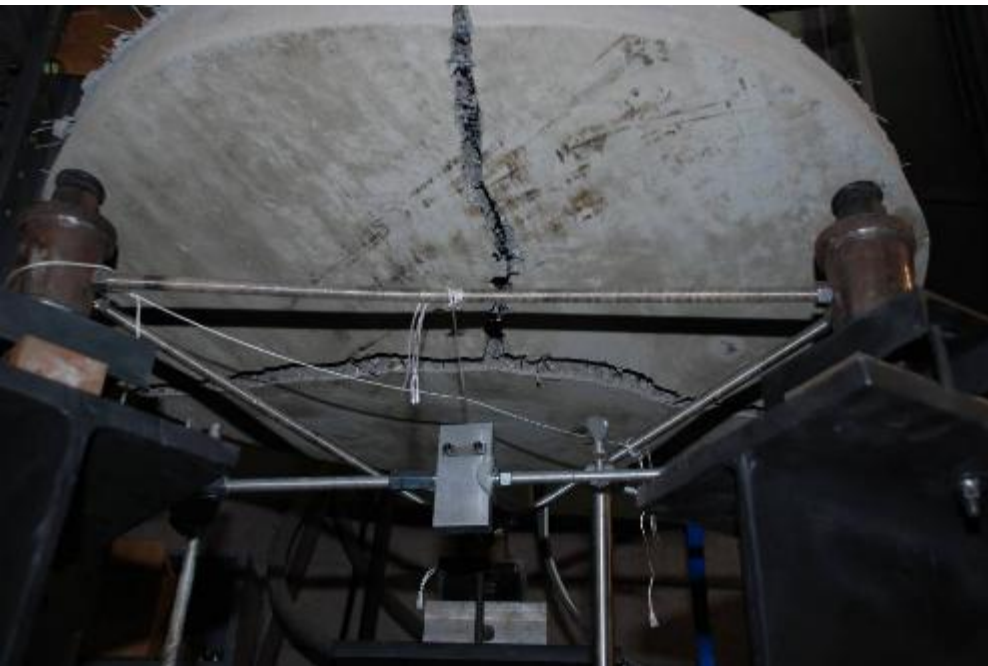
Determinate Round Panel Tests for material modeling



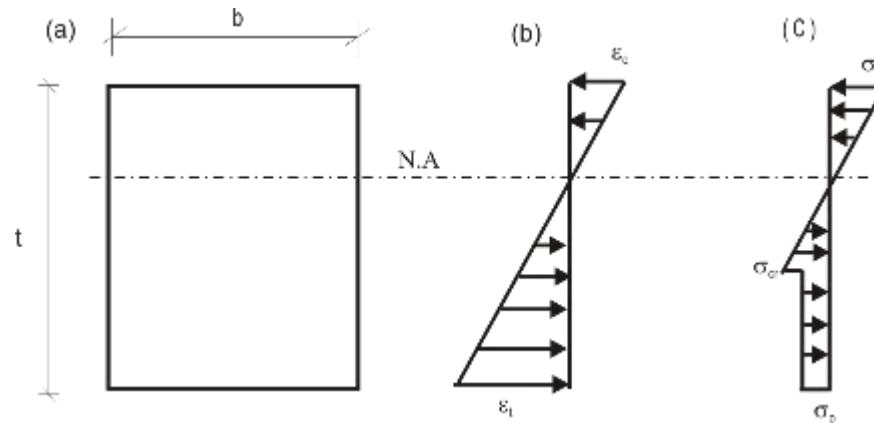
ASTM C1550- Round Panel Test, (RPT)

Load Deflection results used for back-calculation of properties

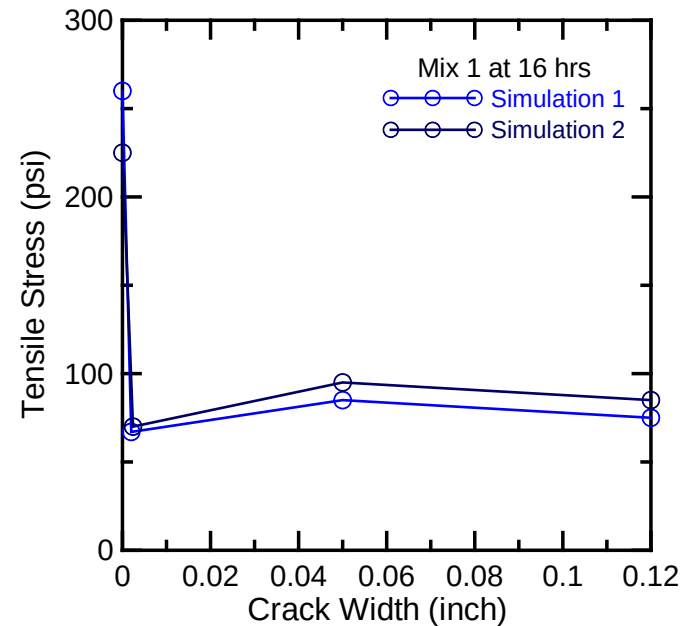
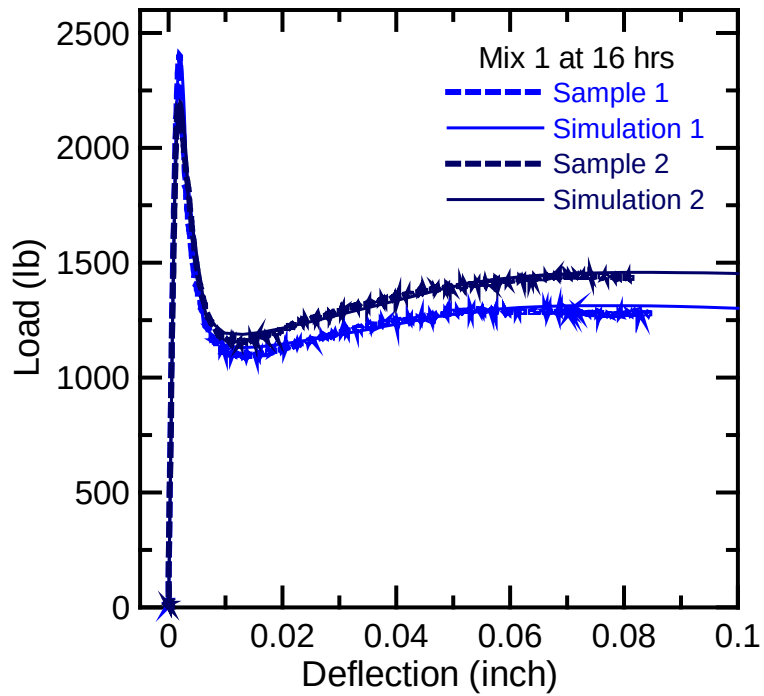




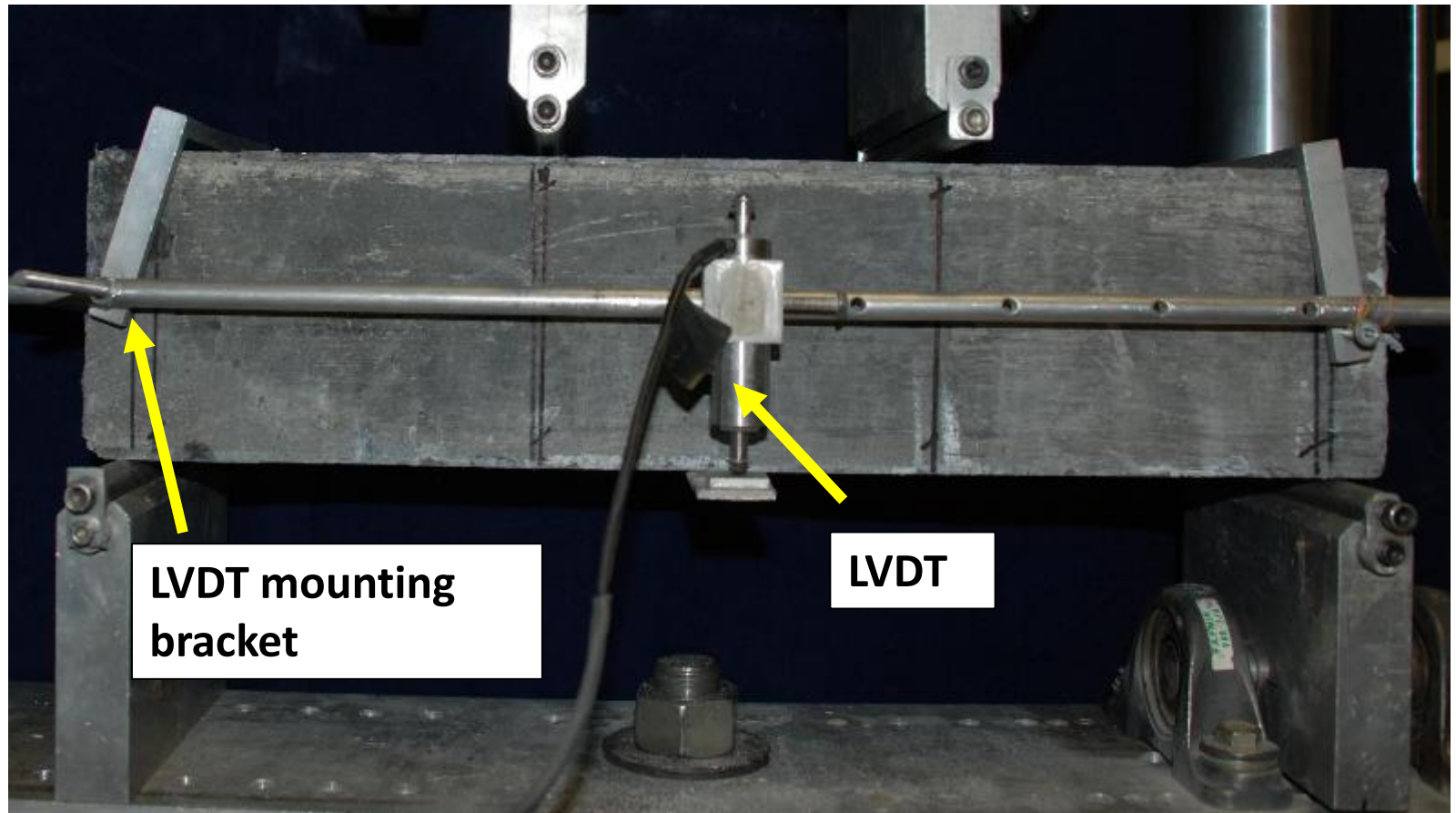
Measurement of residual tensile strength offered by the fibers



Strain and Stress Distribution



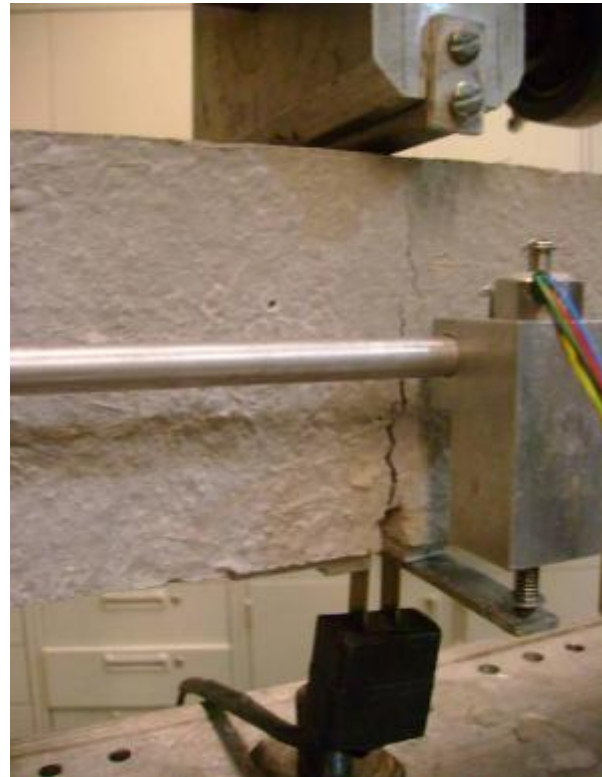
ASTM C1609, flexural beam



Three Point Bending, EN14651 Test notched beam

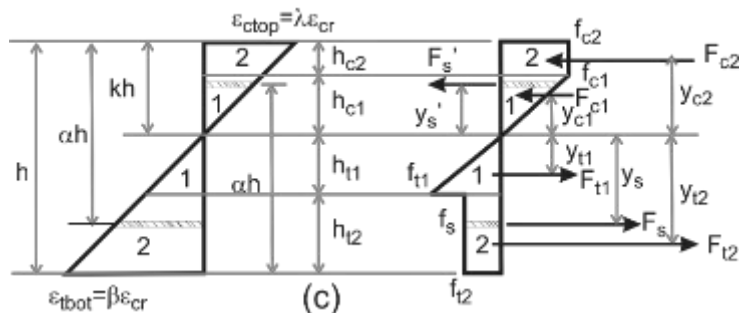
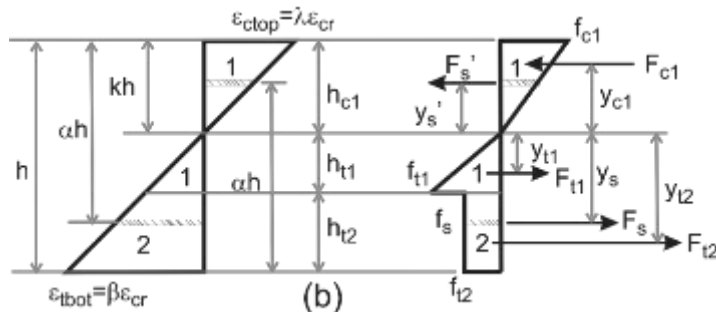
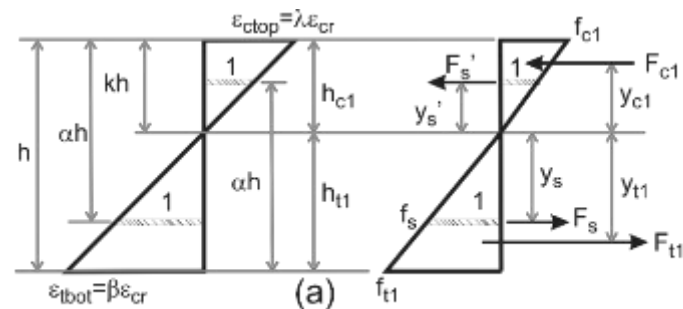
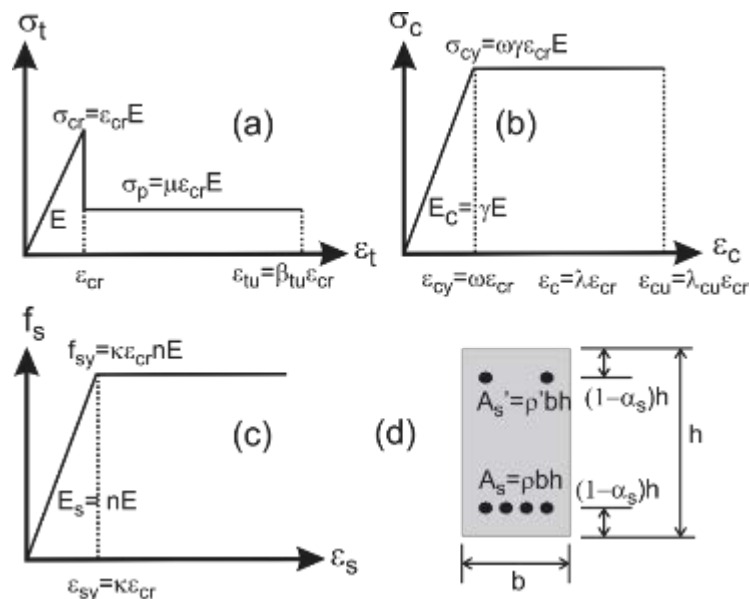


Closed-loop flexural test set up



Crack pattern in a fiber reinforced specimen developed from the notch through concrete mass

Strain- Softening Material Models and Cross-Sectional Analysis



Stress diagrams

Material models

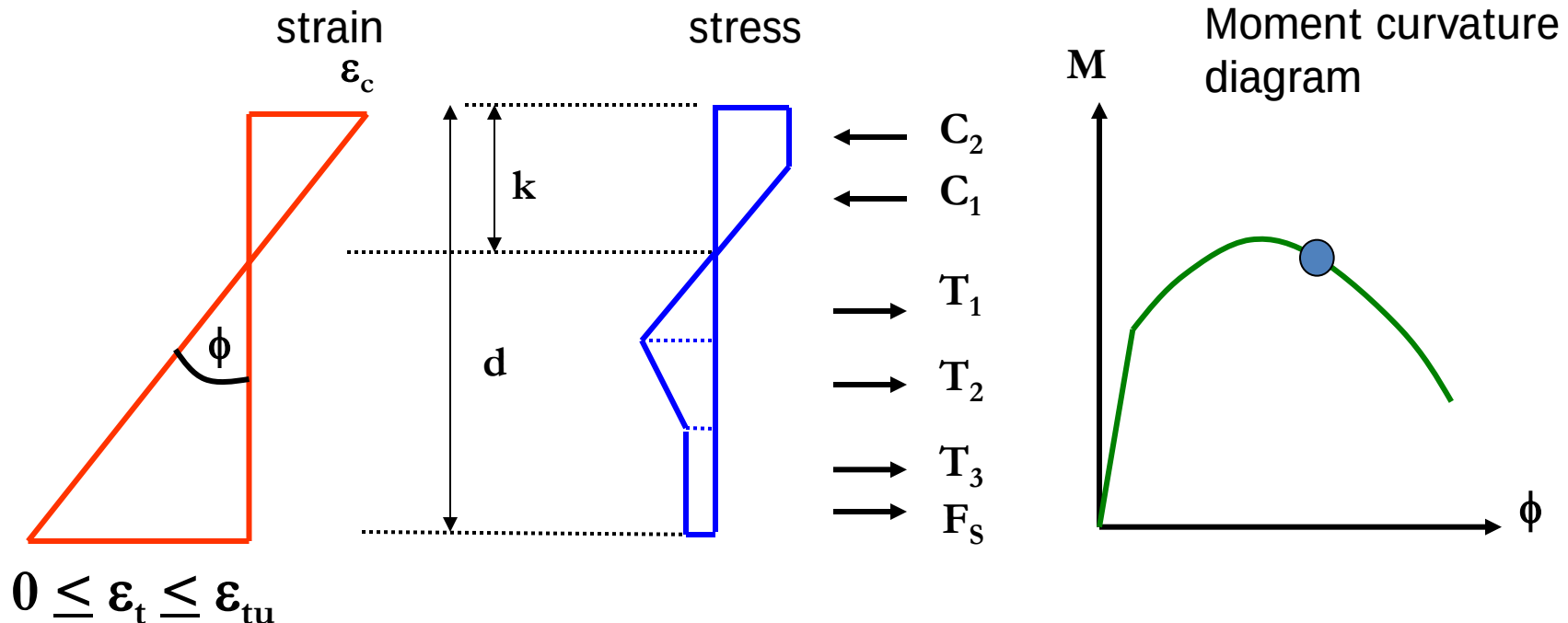
- Plane section remain plane
- Force equilibrium
- Summation of moment
- Moment-curvature response
- Load-deflection response

Moment-Curvature Diagram

- Incrementally impose $0 \leq \varepsilon_t \leq \varepsilon_{tu}$
- Strain Distribution
- Stress Distribution
- $\Sigma F = 0$, determine k (Neutral axis)
- $M = \Sigma C_i y_{ci} + \Sigma T_i y_{ti}$ and $\phi = \varepsilon_c / kd$
- Normalization $M' = M/M_0$ and $\phi' = \phi/\phi_{cr}$

$$F_{c1} = b \int_0^{kd} f_{c1}(y) dy$$

$$y_{c1} = \frac{b}{F_{c1}} \int_0^{kd} f_{c1}(y) y dy$$



Analytical solutions of neutral axis, moment, curvature and stiffness

Stage	k	M'	ϕ'	K'
1	$k_1 = \frac{18\rho_g\alpha + 1}{18\rho_g + 2}$	$M'_1 = \frac{2\lambda}{k_1} [C_1k_1^3 + C_2k_1^2 + C_3k_1 + C_4]$	$\phi'_1 = \frac{\lambda}{2k_1}$	$K'_1 = \frac{M'_1}{\phi'_1}$
2.1	$k_{21} = \frac{\lambda}{B_1} (B_2 + \sqrt{B_3 + 2\alpha\rho_g n B_1})$	$M'_{21} = \frac{1}{\lambda^2 k_{21}} [C_5k_{21}^3 + C_6k_{21}^2 + C_7k_{21} + C_8]$	$\phi'_{21} = \frac{\lambda}{2k_{21}}$	$K'_{21} = \frac{M'_{21}}{\phi'_{21}}$
2.2	$k_{22} = \frac{B_4}{B_1}$	$M'_{22} = \frac{1}{\lambda^2} [C_5k_{22}^2 + C_9k_{22} + C_{10}]$	$\phi'_{22} = \frac{\lambda}{2k_{22}}$	$K'_{22} = \frac{M'_{22}}{\phi'_{22}}$
3.1	$k_{31} = \frac{\lambda}{B_5} (B_2 + \sqrt{B_3 + 2\alpha\rho_g n B_5})$	$M'_{31} = \frac{1}{\lambda^2 k_{31}} [C_{11}k_{31}^3 + C_6k_{31}^2 + C_7k_{31} + C_8]$	$\phi'_{31} = \frac{\lambda}{2k_{31}}$	$K'_{31} = \frac{M'_{31}}{\phi'_{31}}$
3.2	$k_{32} = \frac{B_4}{B_5}$	$M'_{32} = \frac{1}{\lambda^2} [C_{11}k_{32}^2 + C_9k_{32} + C_{10}]$	$\phi'_{32} = \frac{\lambda}{2k_{32}}$	$K'_{32} = \frac{M'_{32}}{\phi'_{32}}$

where, the coefficients are:

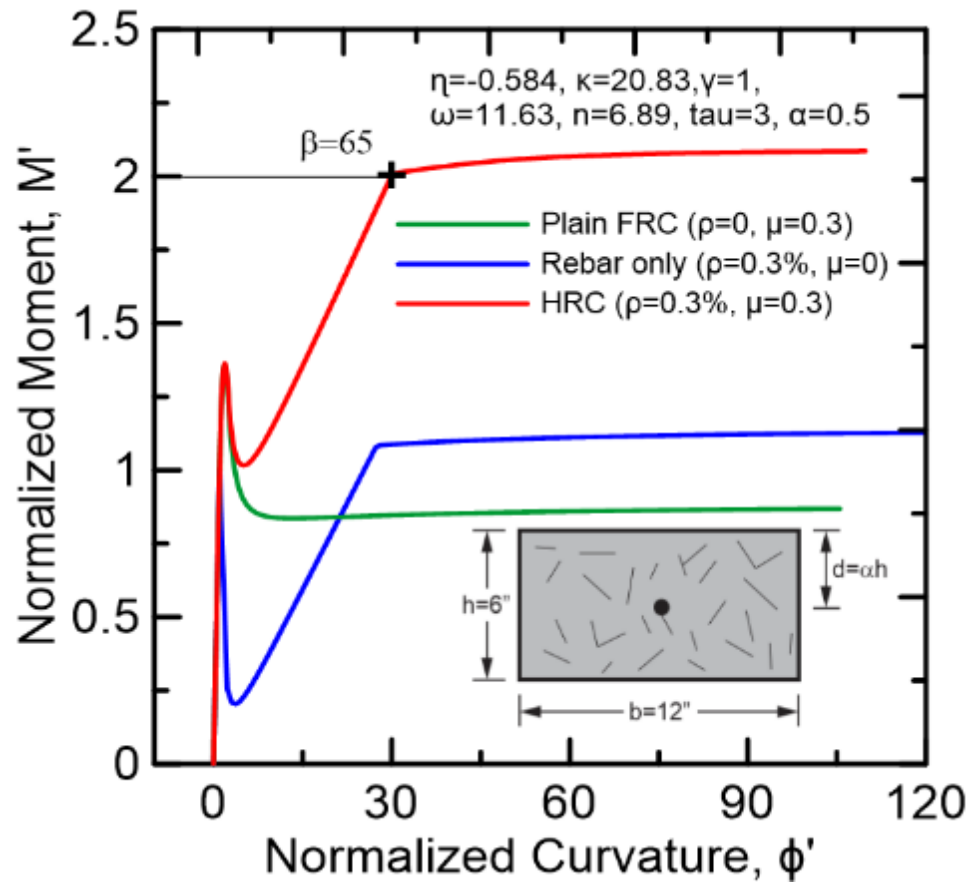
$$\begin{aligned}
 B_1 &= \lambda^2 + 2\mu(\lambda + 1) - 1; \quad B_2 = \mu - 9\rho_g\lambda; \quad B_3 = 9\rho_g(\rho_g 9\lambda^2 - 2\mu\lambda) + \mu^2; \quad B_4 = 2\lambda(9\rho_g\kappa + \mu); \quad B_5 = 20\lambda - 101 + 2\mu(\lambda + 1) \\
 C_1 &= 0; \quad C_2 = 27\rho_g + 3; \quad C_3 = -3 - 54\rho_g\alpha; \quad C_4 = 1 + 27\rho_g\alpha^2; \quad C_5 = 2\lambda^3 + 3\mu(\lambda^2 - 1) + 2; \quad C_6 = 6\lambda^2(9\lambda\rho_g - \mu) \\
 C_7 &= 3\lambda^2(\mu - 36\rho_g\alpha\lambda); \quad C_8 = 54\rho_g\alpha^2\lambda^3; \quad C_9 = -6\lambda^2(9\rho_g\kappa + \mu); \quad C_{10} = 3\lambda^2(18\rho_g\alpha\kappa + \mu); \quad C_{11} = 30\lambda^2 + 3\mu(\lambda^2 - 1) - 998
 \end{aligned}$$

Modeling and Validation of Experimental approach

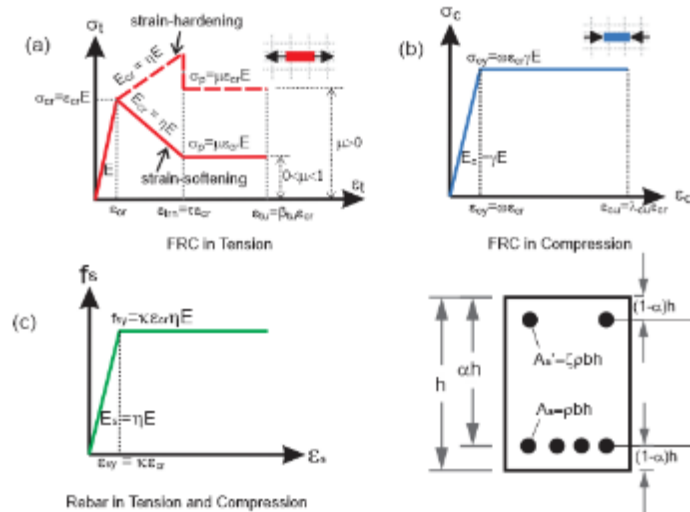
- Hybrid Structural design: First principal tools using to incorporate hybrid Supplementary Reinforcing Materials,
- Labor, Efficiency: Enhancing construction speed, design alternative checks
- Performance-based Metrics: Analysis, Design, and novel test methods
- Design for serviceability and durability: Different reinforcements operate at different strain ranges
- Optimization of generalized solutions -design envelop for new and repair/retrofit apps
- Applicable to:
 - fibers, textiles, FRP, or ordinary steel reinforcement systems.
 - Structural elements meeting durability or serviceability limit states.



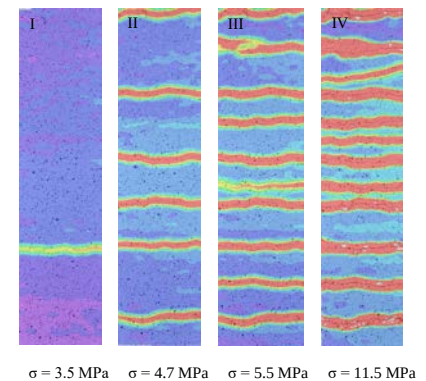
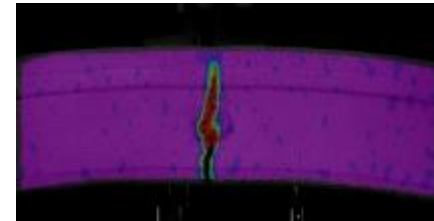
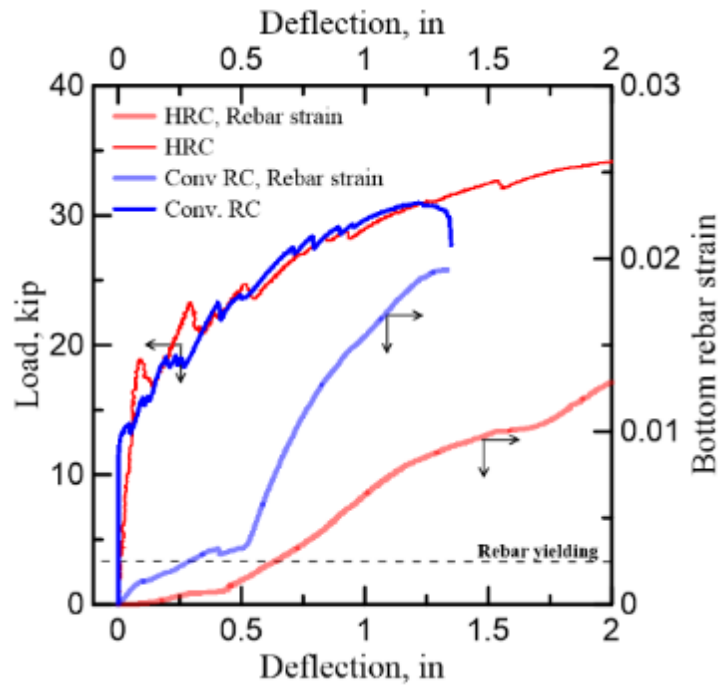
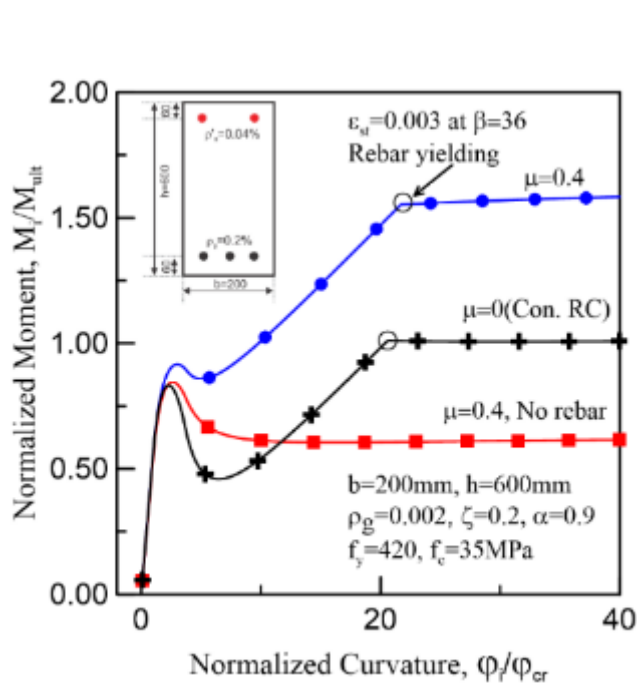
Contribution of Fibers to the reinforced concrete



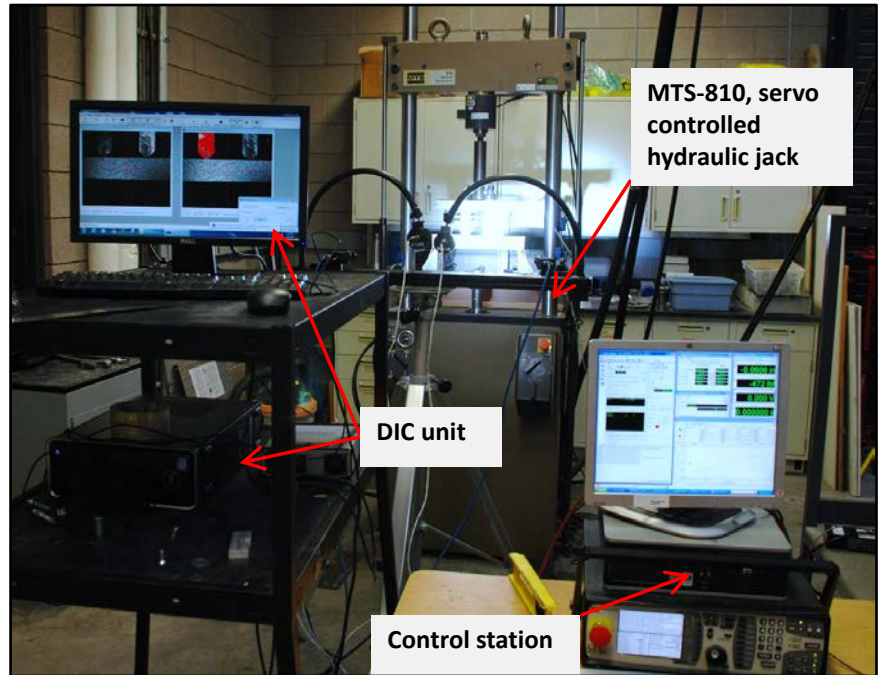
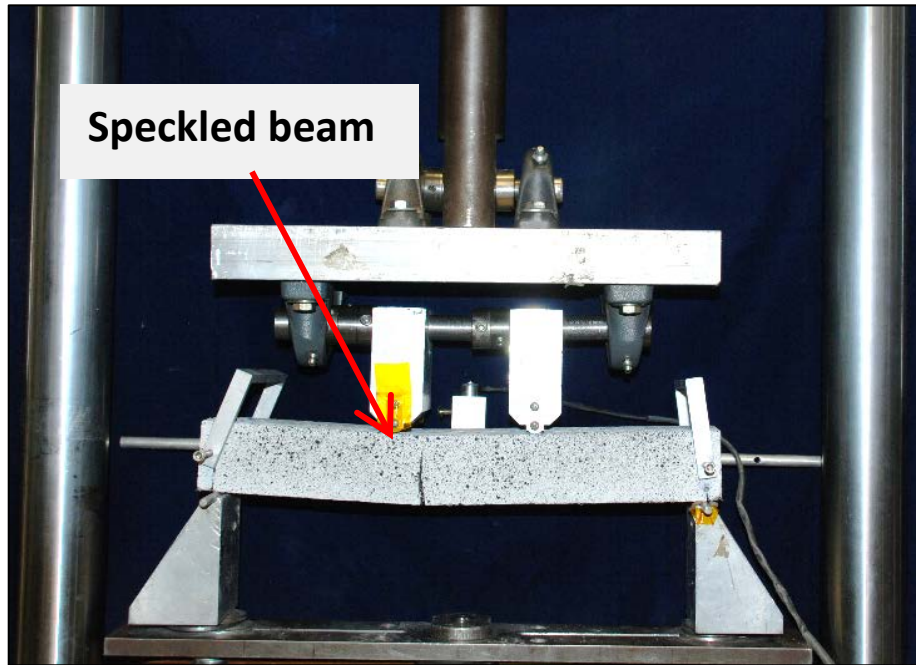
Generalized Moment Curvature Parametric Approach



Analytical and Experimental Approach for Hybrid Design

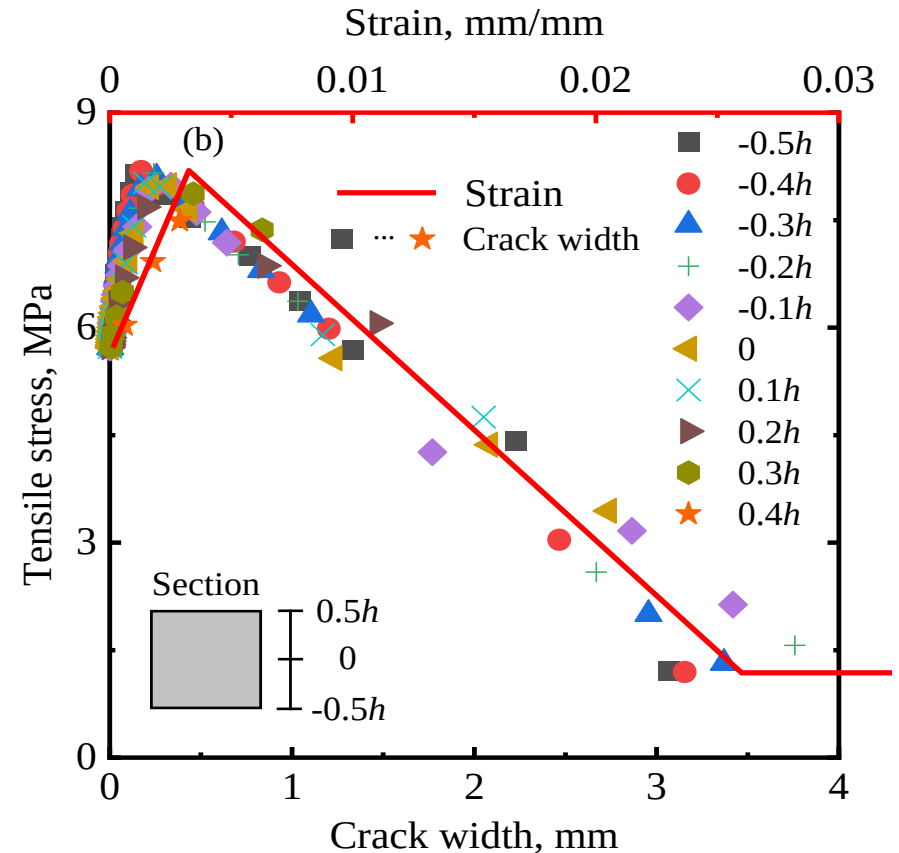
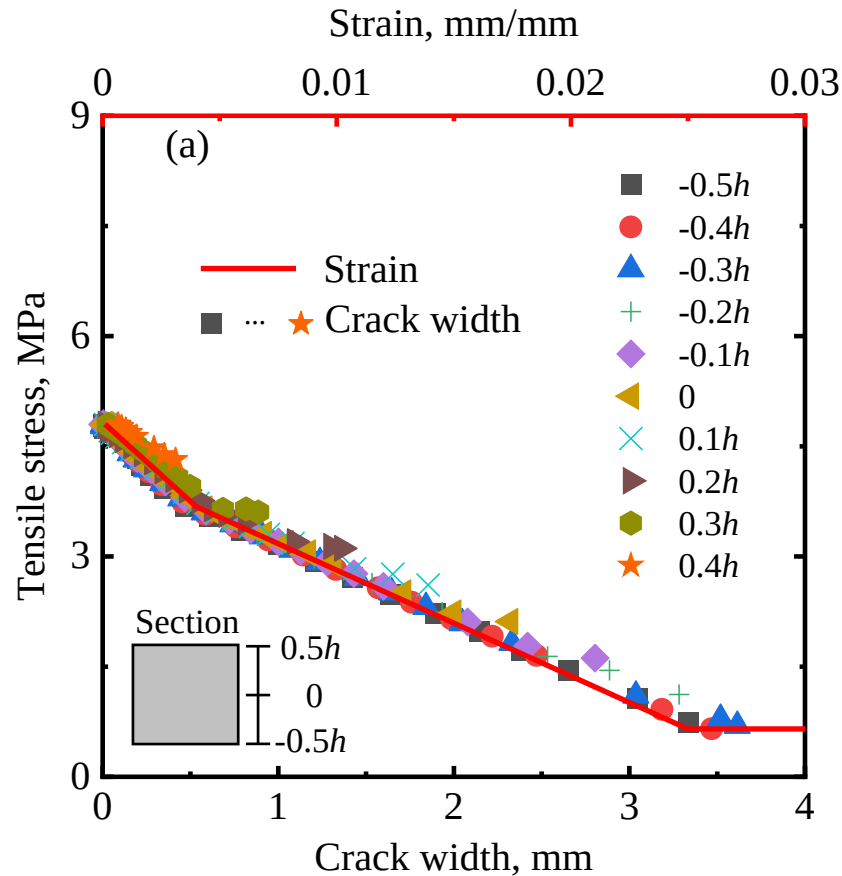


Flexural closed loop tests

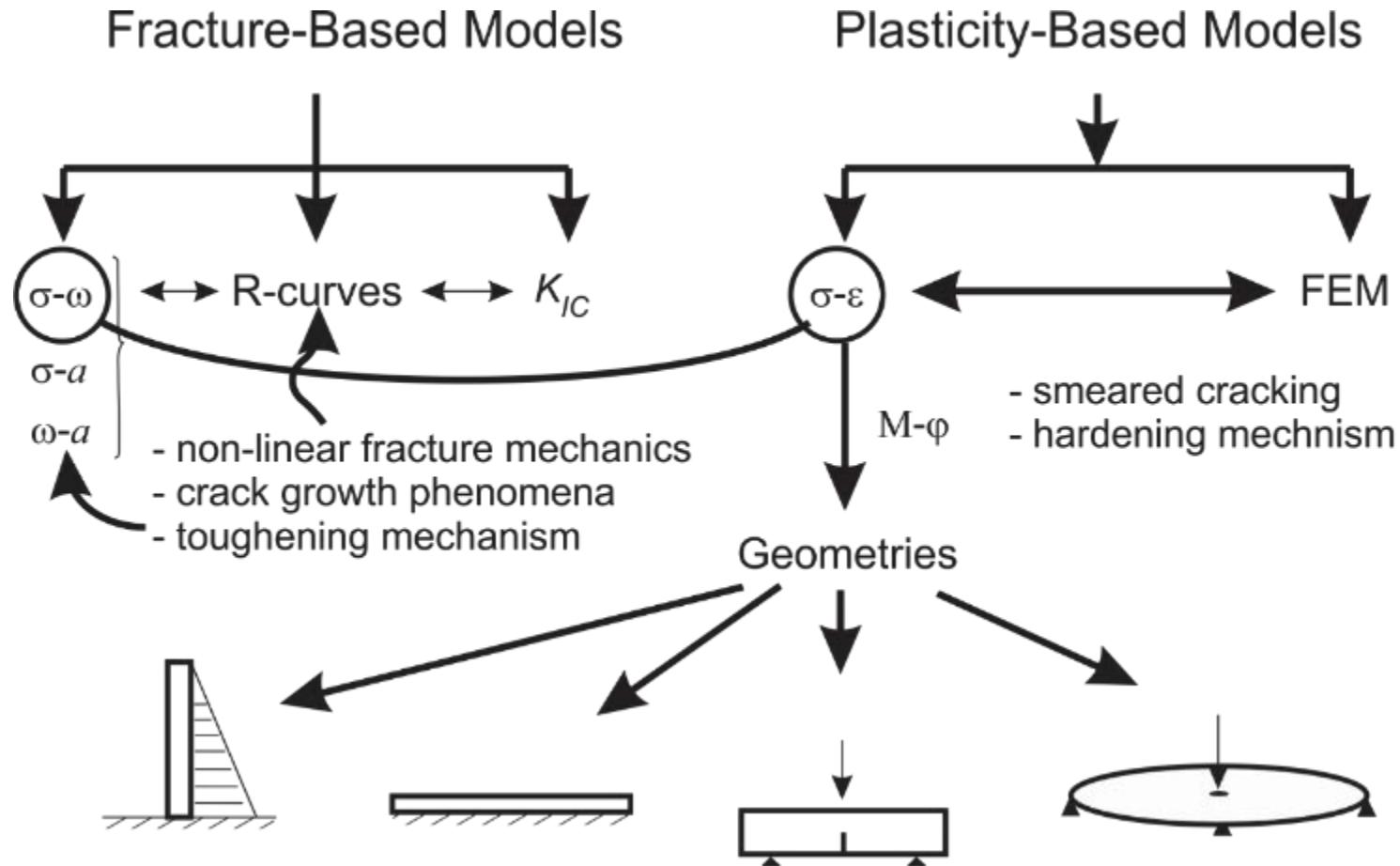


Stress-Crack width Relationship

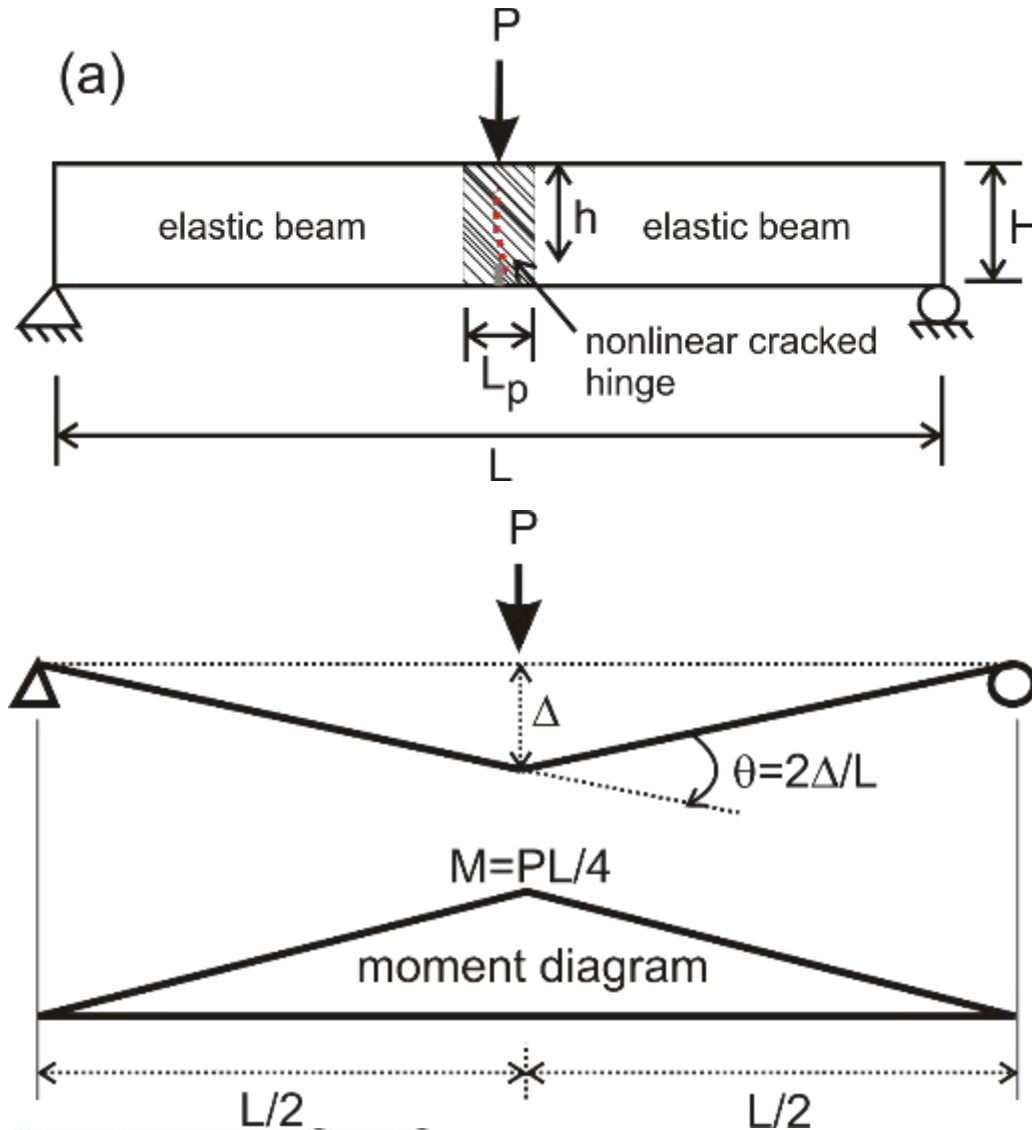
Calibration of model and experimental data



Fracture and plasticity models



Elastic Beam with Nonlinear Cracked Hinge Model



- Assume beam rotates rigidly after cracking

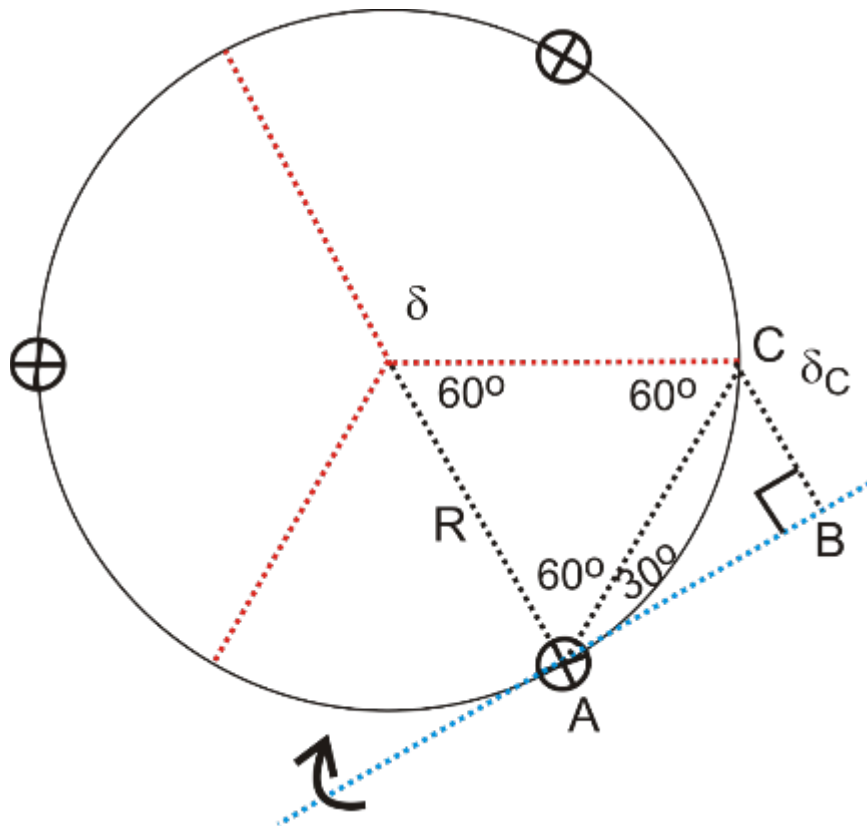
- Rotation
$$\theta = \frac{2\Delta}{L}$$

- Moment
$$M = \frac{PL}{4}$$

- Moment per unit width

$$m = \frac{M}{b}$$

Rigid Cracked Panel Model



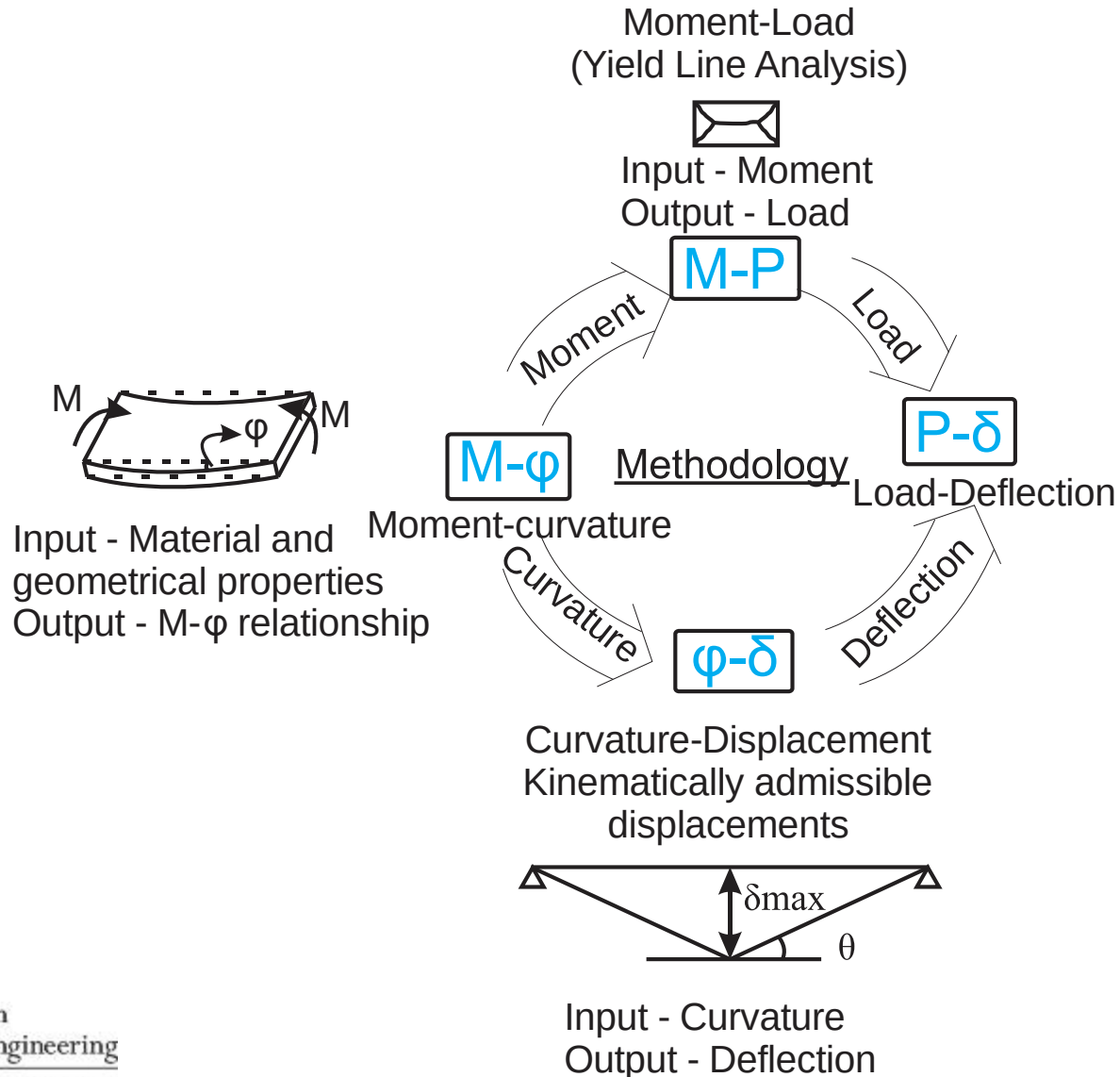
- Assume panel rotates rigidly after cracking
- Relationship between rotation and deflection

$$\theta = \cos^{-1} \left\{ \frac{-\delta^2 + 2R^2}{2(\delta^2 + R^2)} \right\}$$

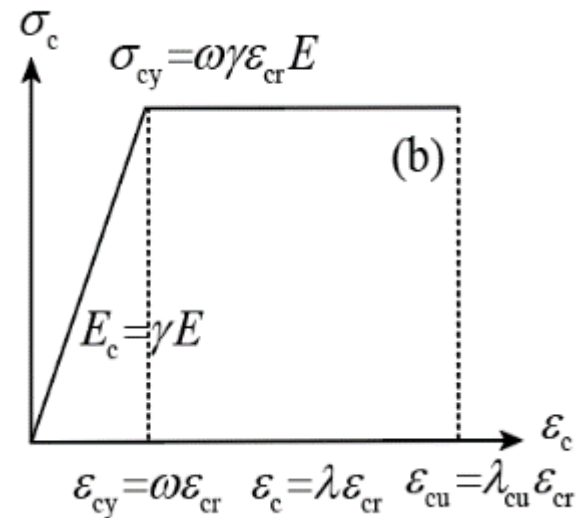
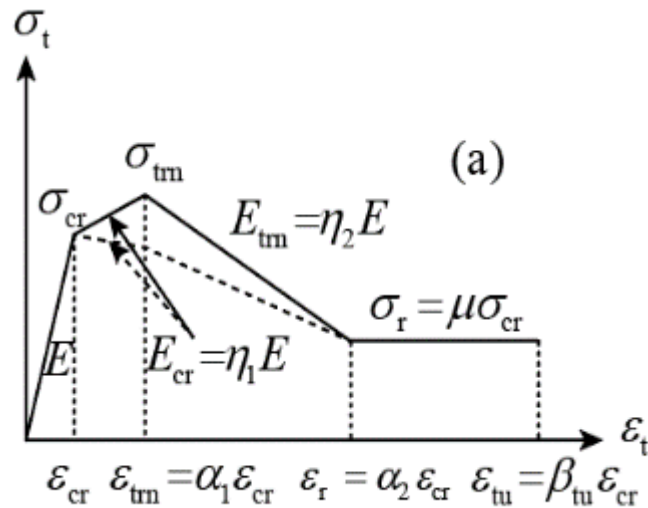
- By yield line analysis, the load depends on plastic energy along the three cracked lines.

$$P = \frac{3Rm\theta}{\delta}$$

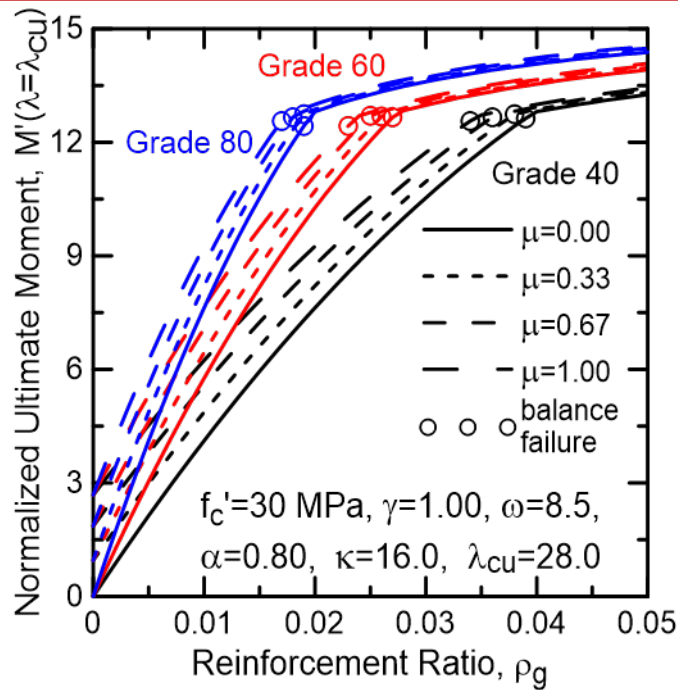
Generalized Modeling



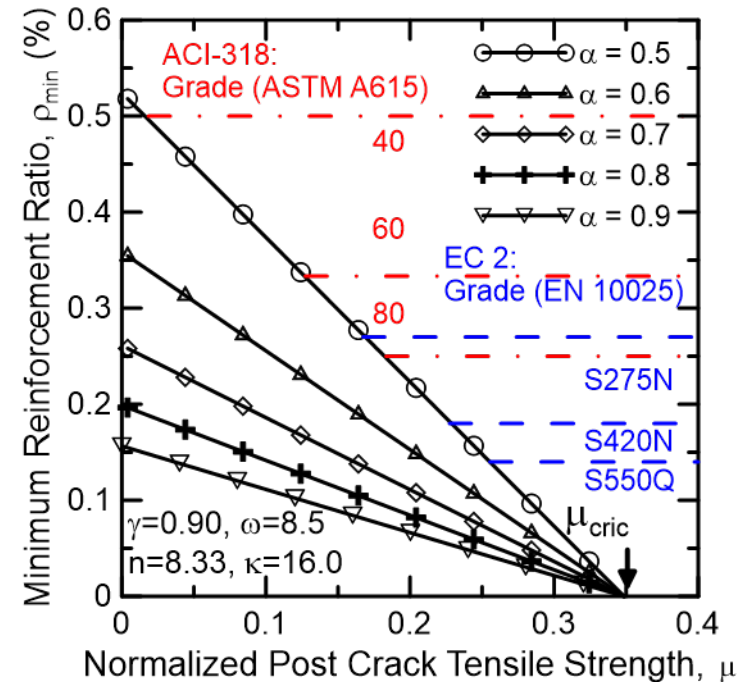
Quad-Linear based Model development for Strain Hardening Materials



Design Charts: Moment Capacity and Minimum Reinforcement Ratio



$$M_u \approx M'_\infty M_{cr} = \frac{6\rho_g n\kappa(\mu\alpha - \mu + \alpha\omega) + 3\omega\mu - 3(\rho_g n\kappa)^2}{\omega + \mu} M_{cr}$$



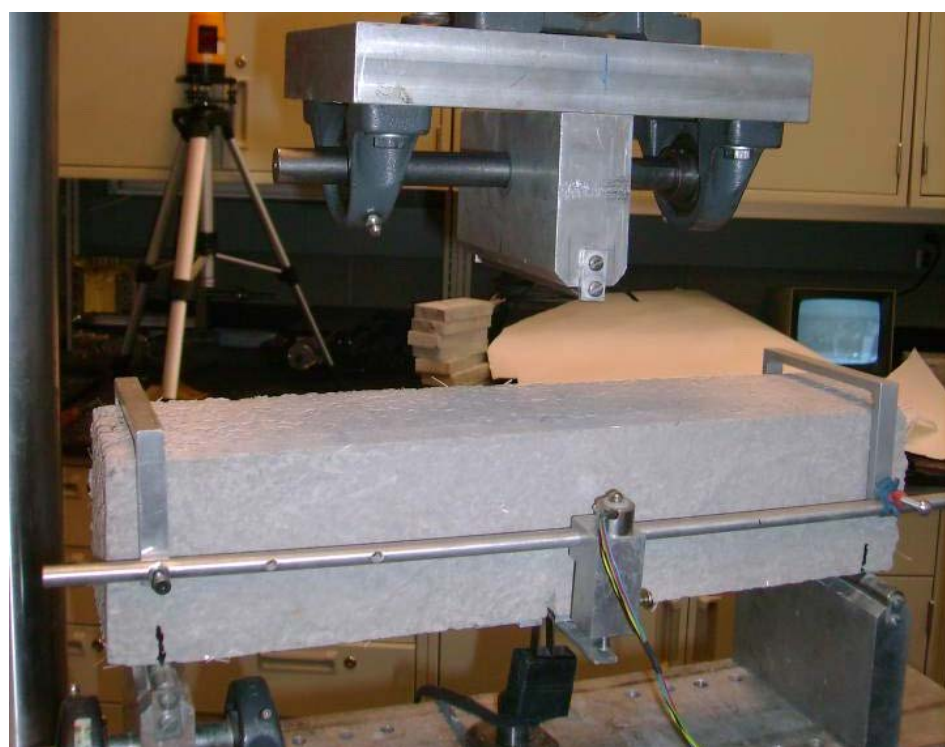
$$\rho_{min} = \frac{3G_1 - \sqrt{G_2 + 9G_1^2}}{3\alpha n\kappa}$$

- Role of steel fiber in reducing the longitudinal reinforcements is revealed
- Parametric design charts simplify the proper selection of variables

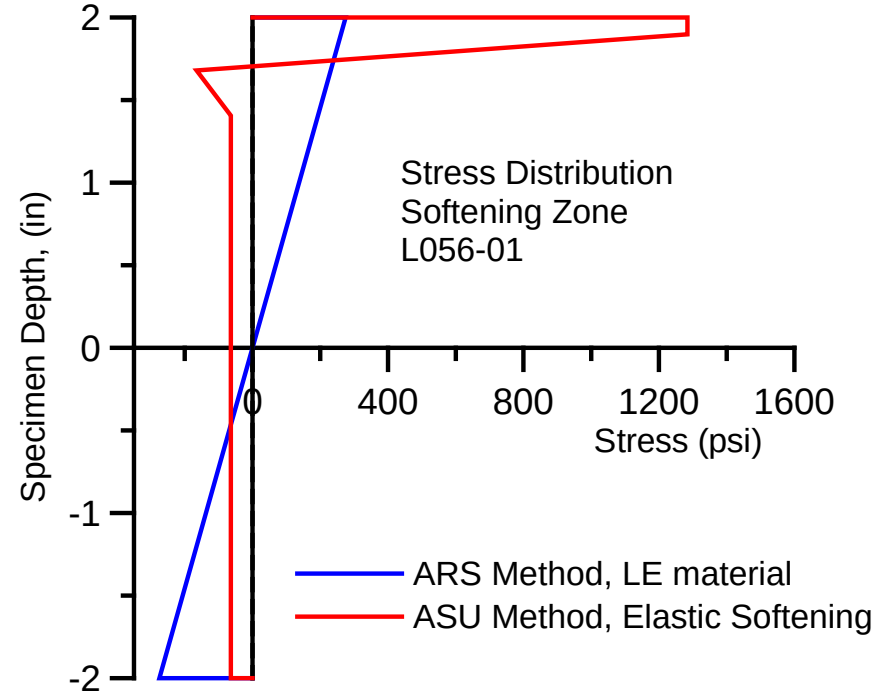
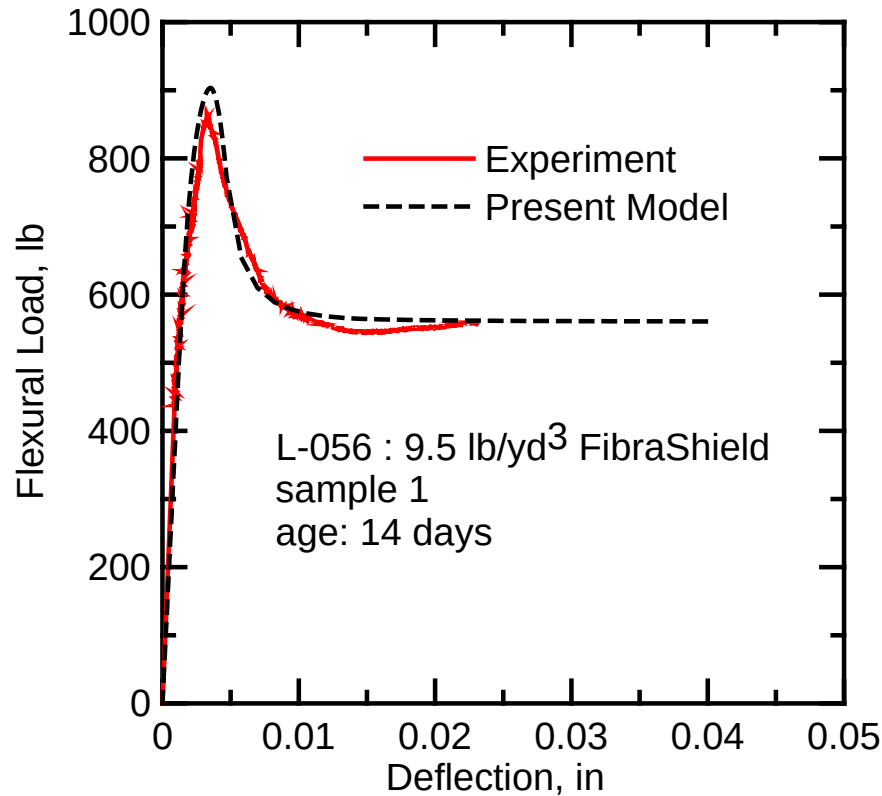
Mobasher, B., Yao, Y., Soranakom, C. (2015). Analytical solutions for flexural design of hybrid steel fiber reinforced concrete beams. *Engineering Structures*, 100, 164-177.

Correlation between Three Point Bending Test and Round Panel Test

- There are two majors testing methods to evaluate post crack performance of fiber reinforced concrete
 - Three-point bending
 - Determinate round panel test



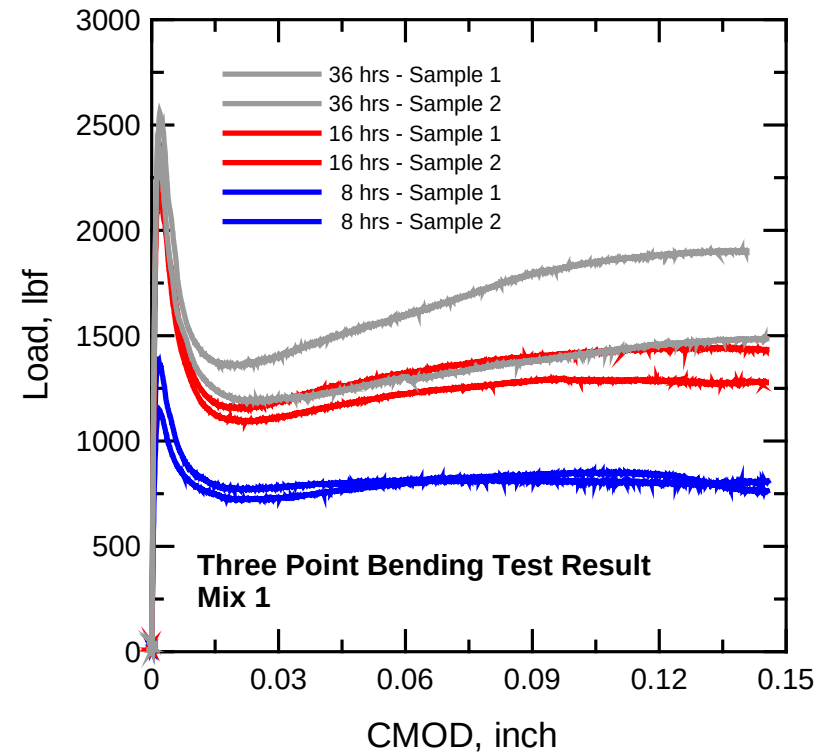
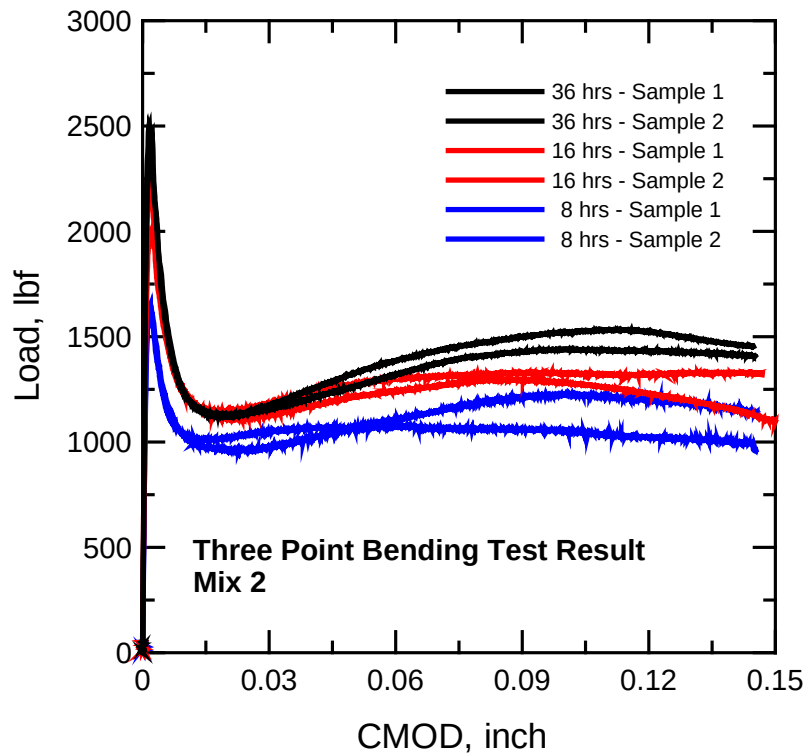
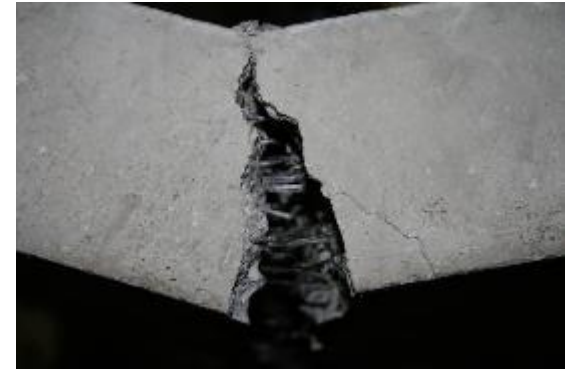
Linear Elastic Analysis will result in misleading results, ASTM- C1609



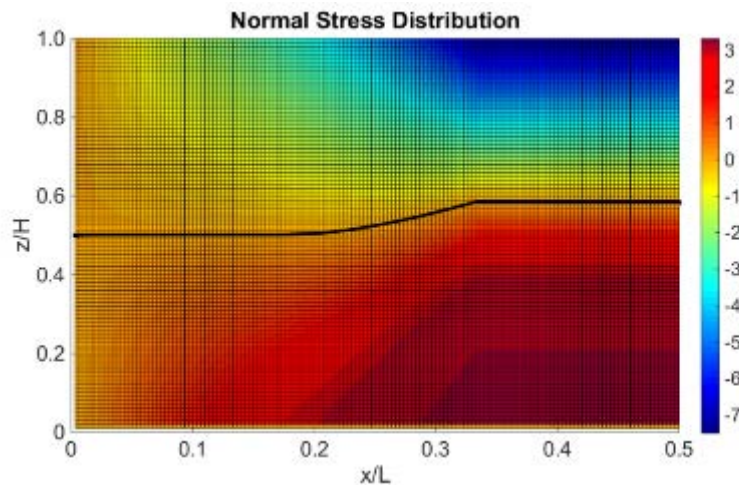
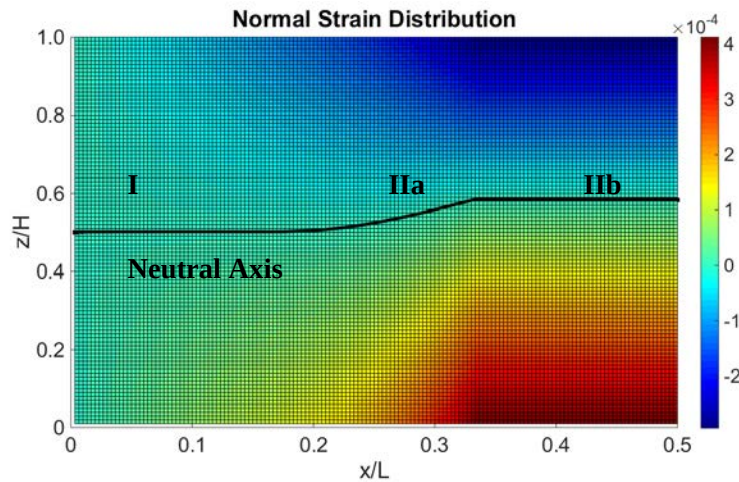
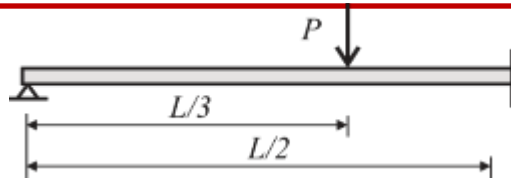
Notched beam test Early age development of flexural response



10-12 lbs/yd³ of macro fibers

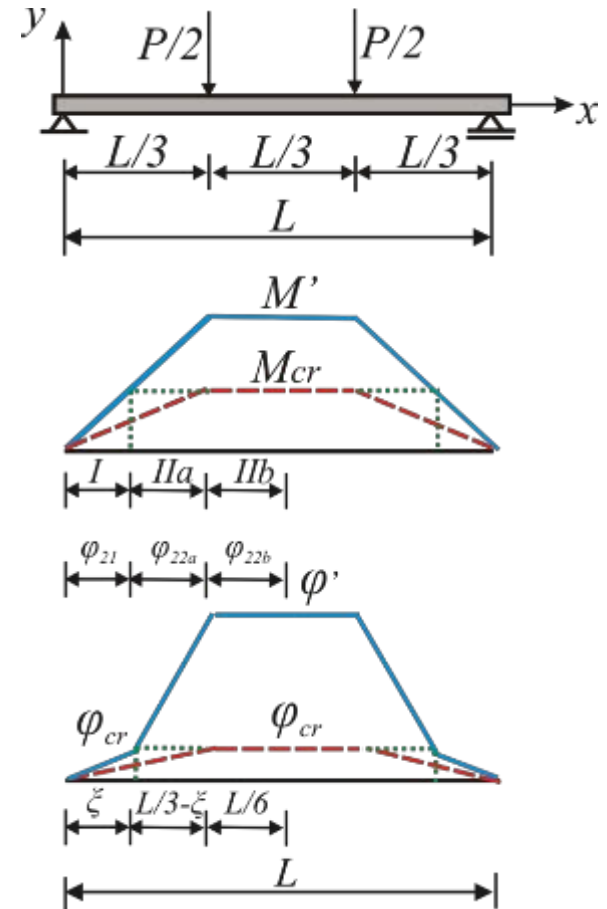


2D Contour – Distribution loading

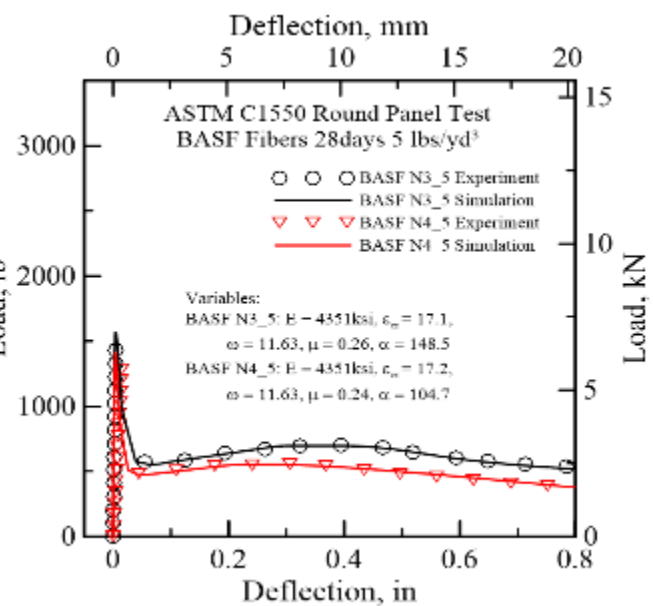
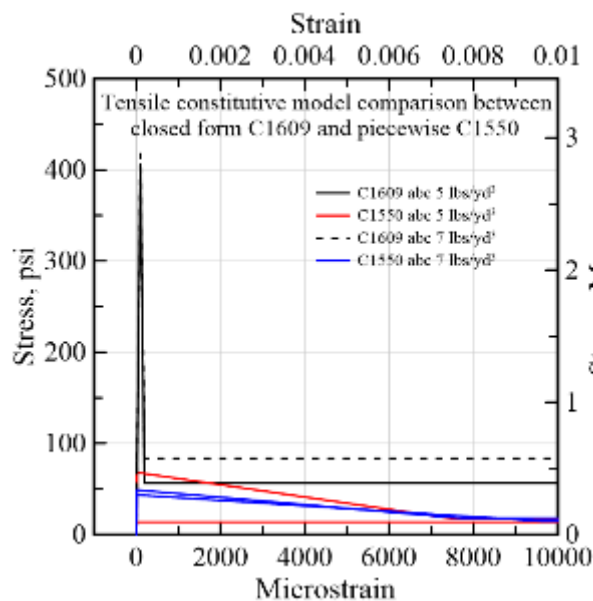
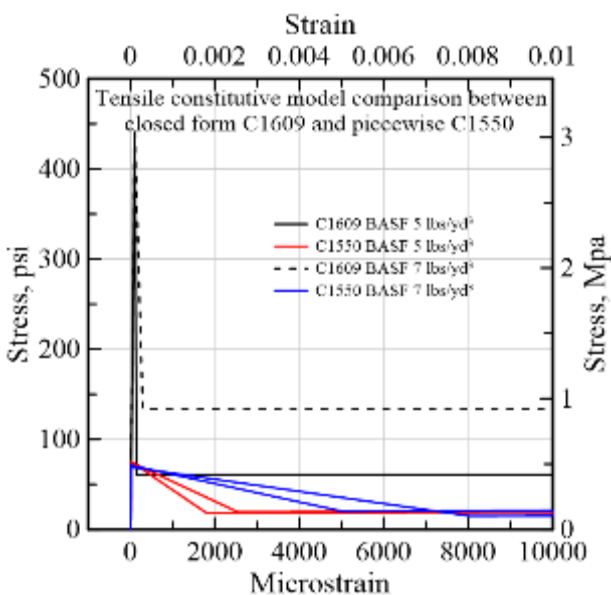
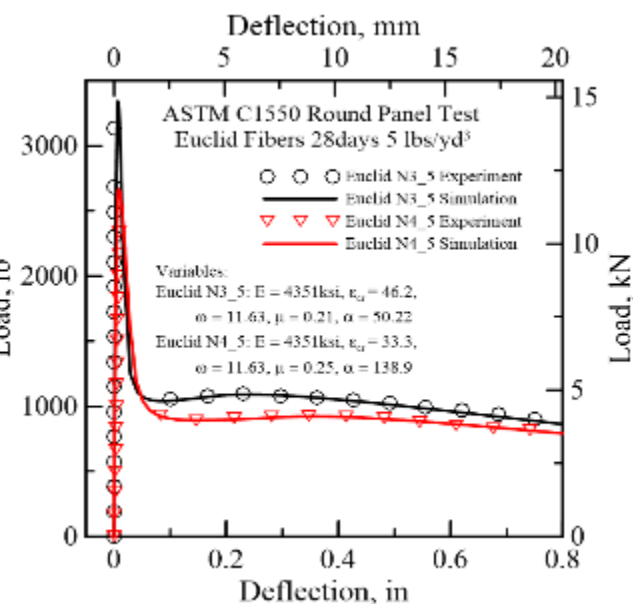
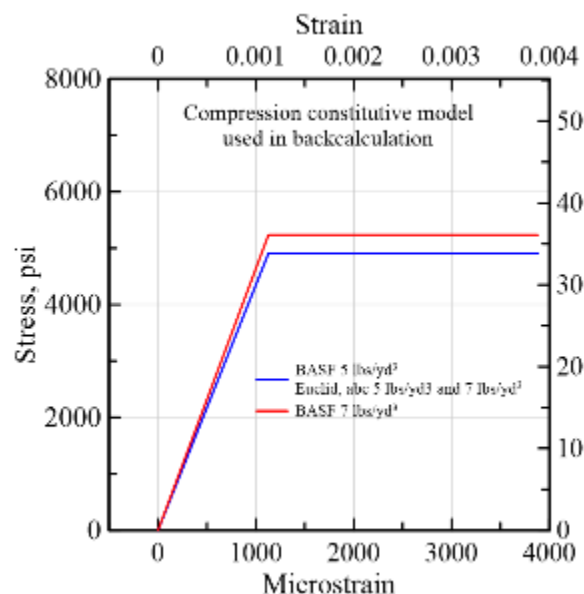
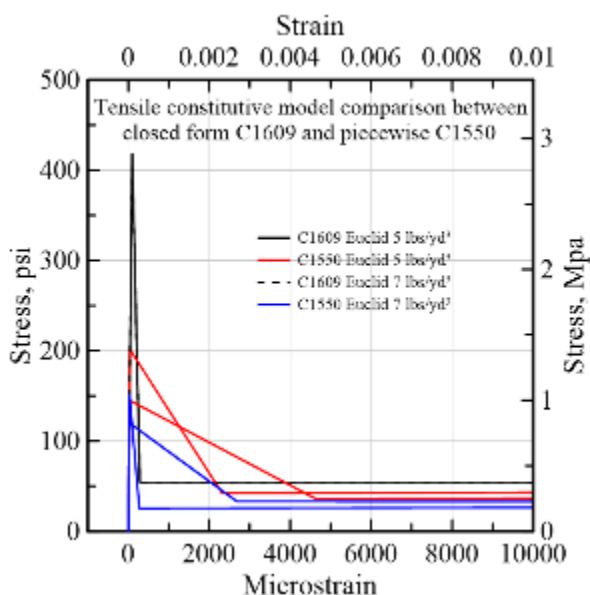


Normal
strain
distribution

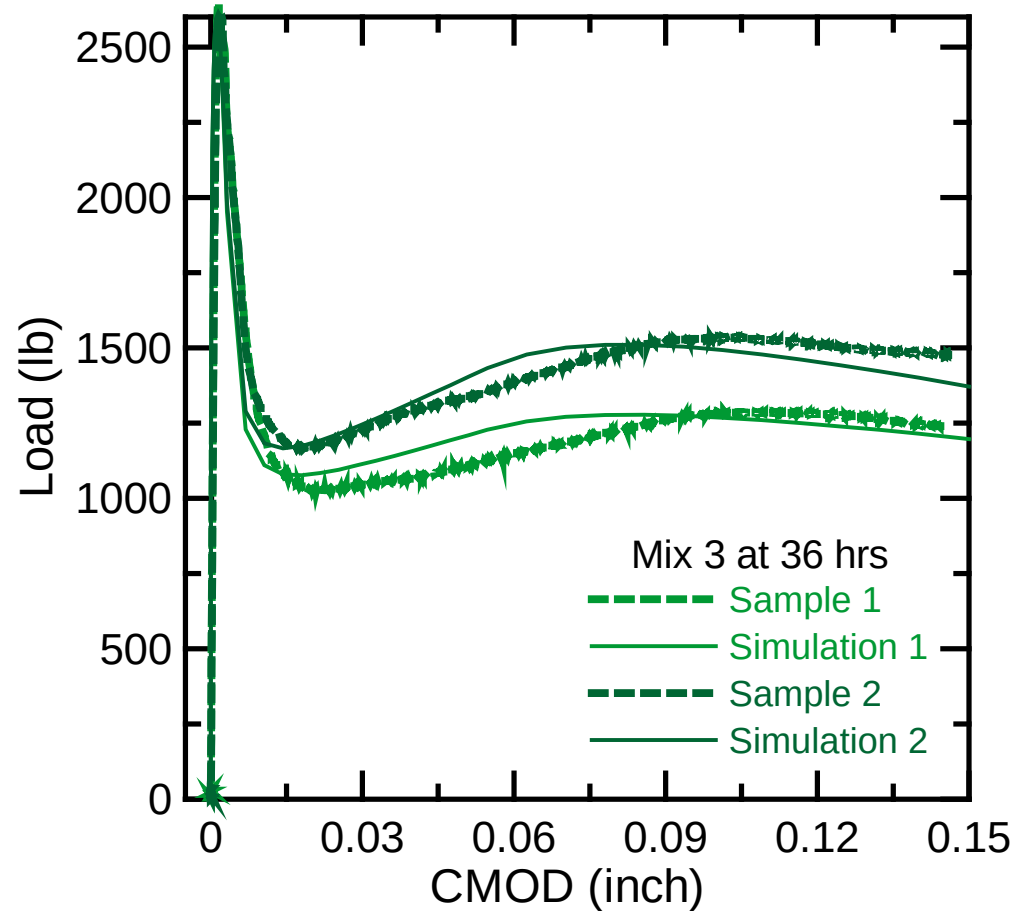
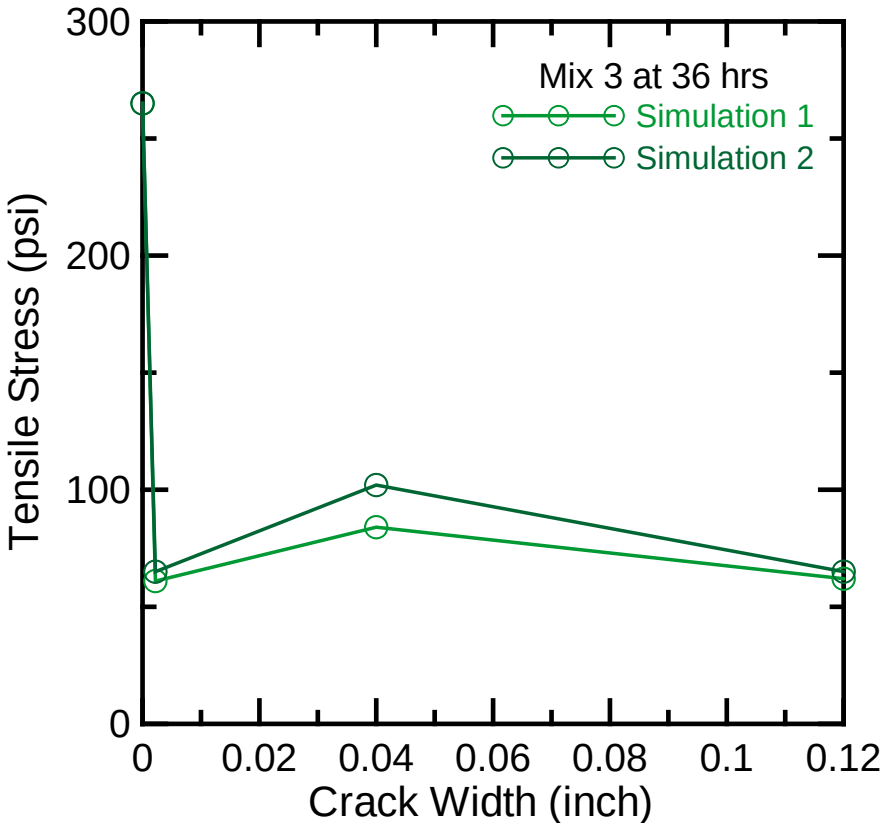
Normal
stress
distribution



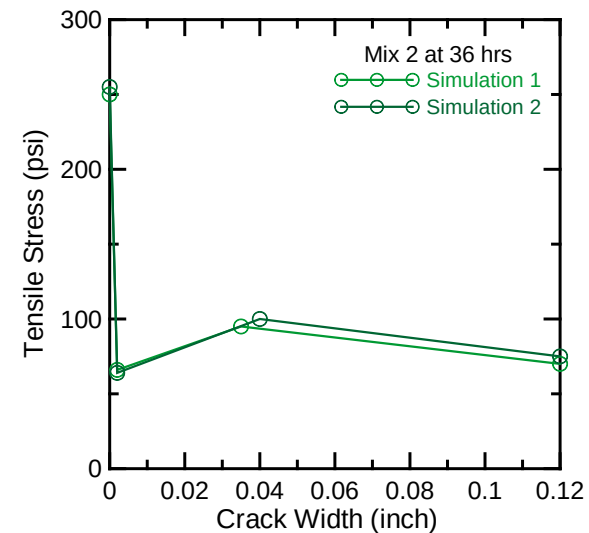
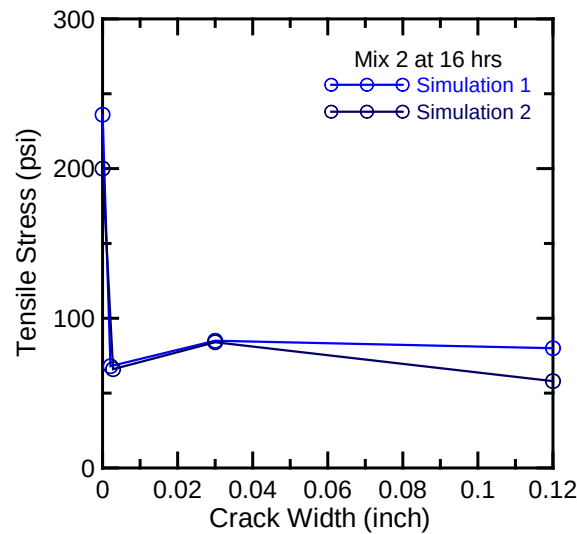
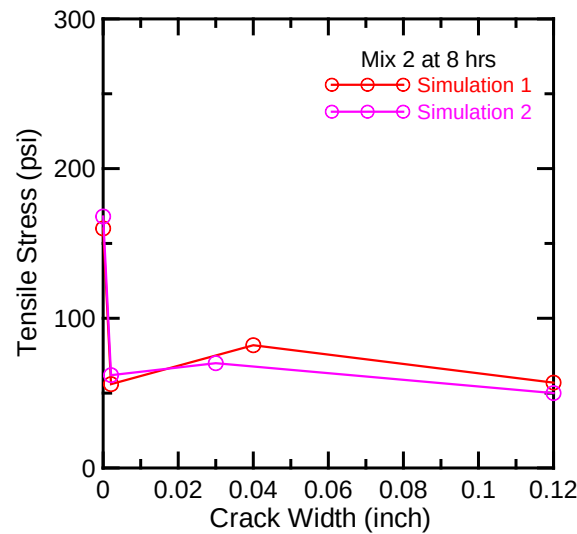
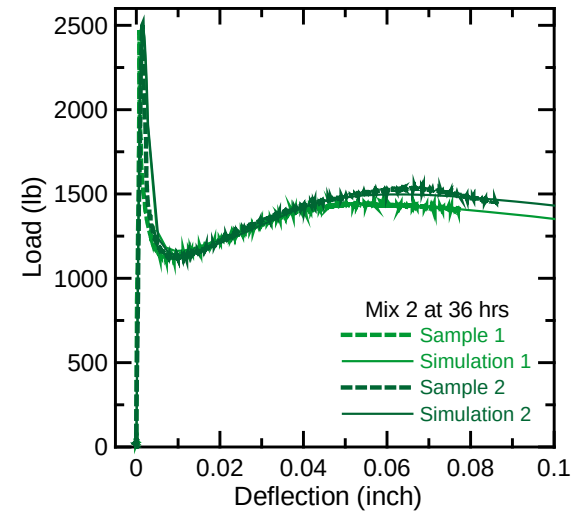
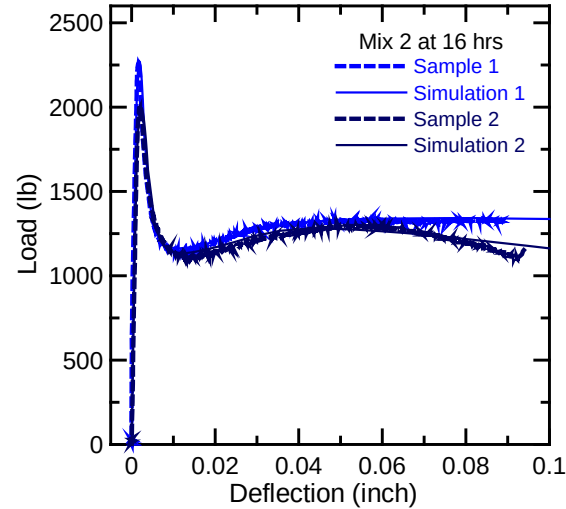
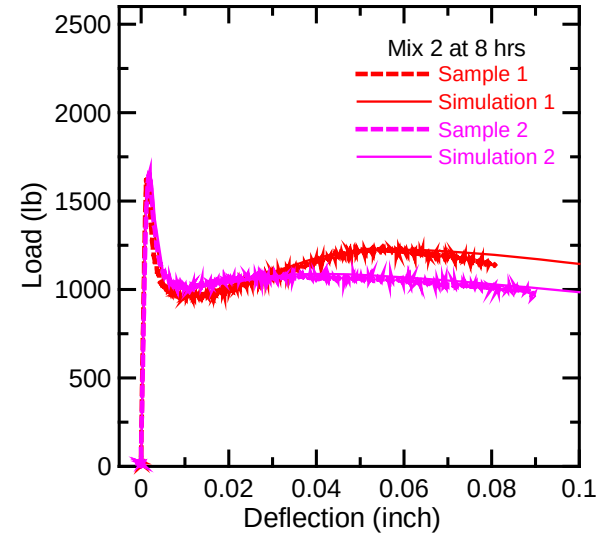
Back-calculated material Parameters and predicted load-deflection results



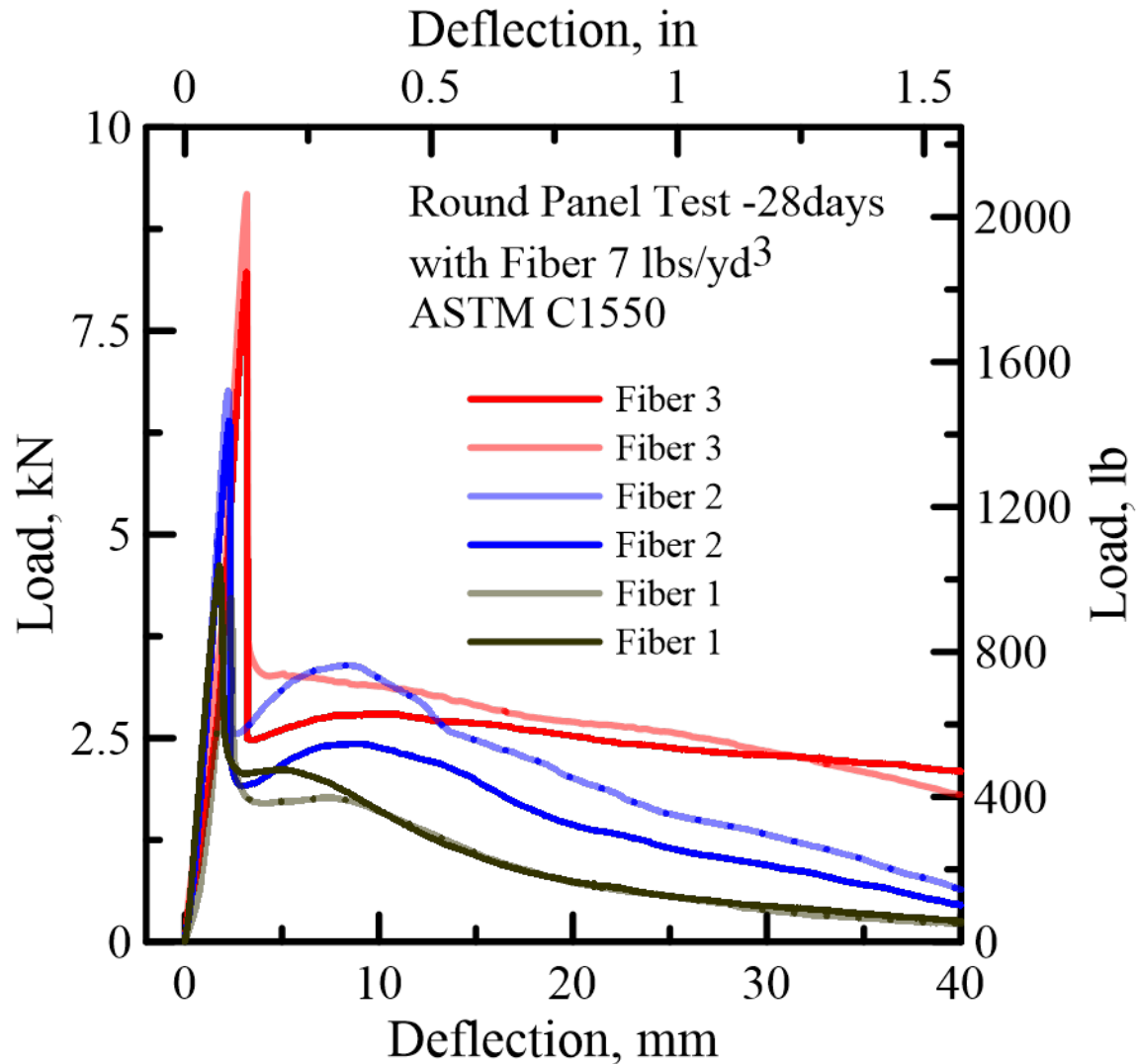
Inverse Analysis Procedures- Macro fiber dosage level 10 lb/yd³



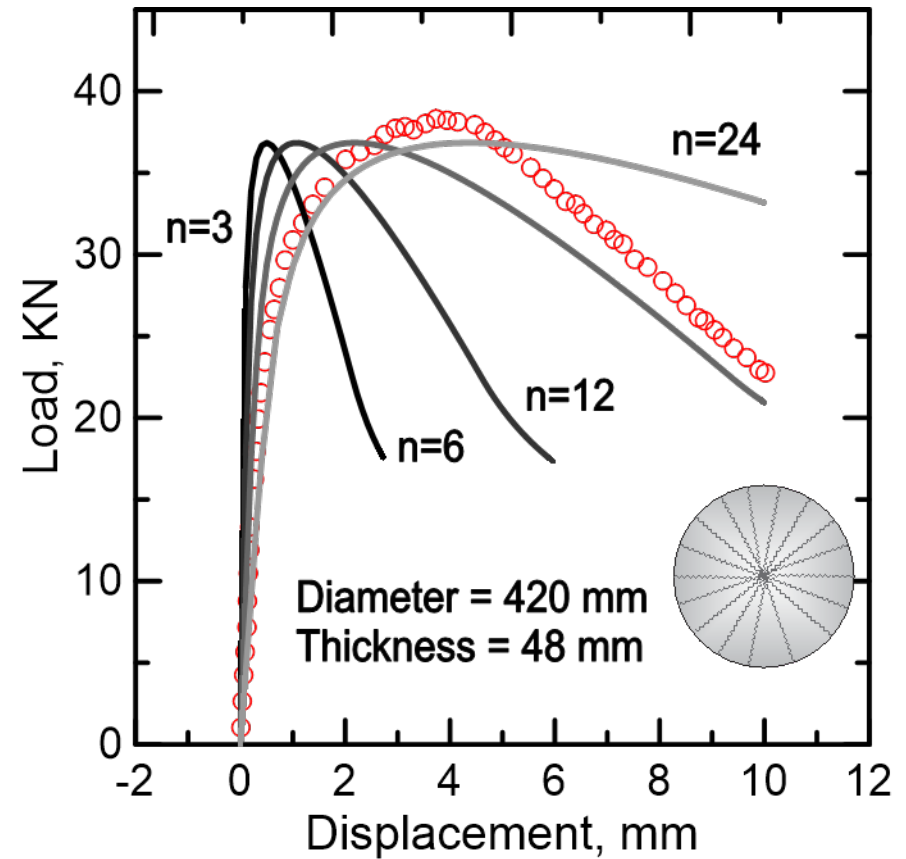
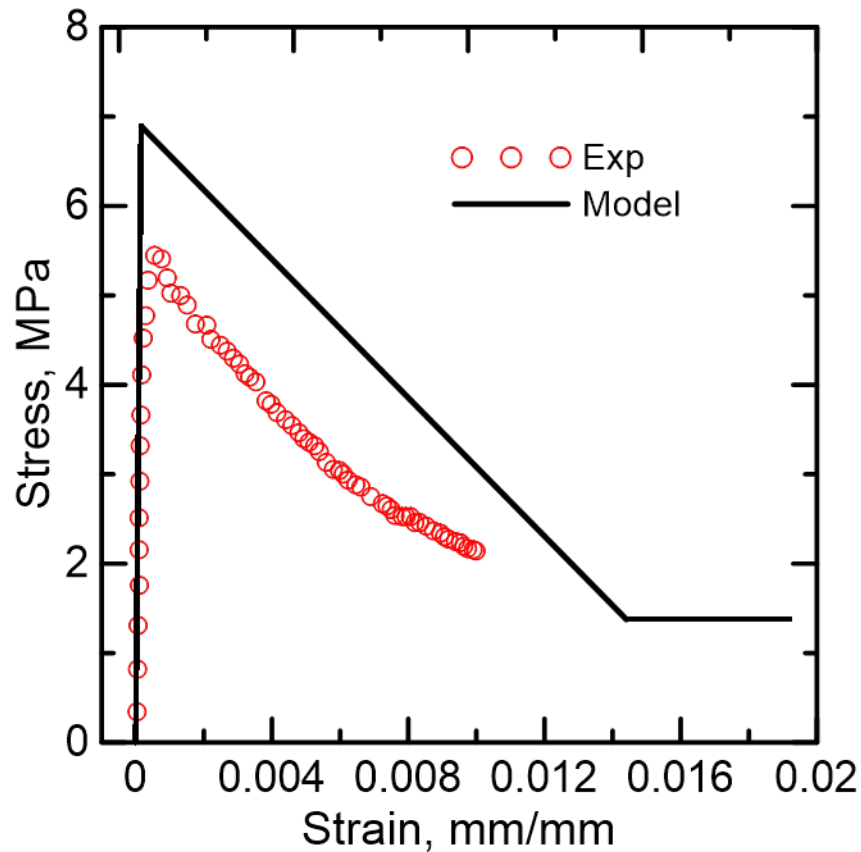
Data Reduction by model fitting



Round Panel Product Evaluation



Effect of Number of Yield Lines





Stress-Strain model to Load-deflection of square slabs

